DTU Compute
Department of Applied Mathematics and Computer Science

Heart and Sole

Insole-Based Ballistocardiography System for Heart Rate Monitoring

Bachelor Thesis

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August 5, 2026

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Cover photo: Vibeke Hempler, 2012
Published by: DTU, Department of Applied Mathematics and Computer Science, Richard Petersens Plads, Building 324, 2800 Kgs. Lyngby Denmark
https://www.compute.dtu.dk/

Approval

This thesis has been prepared over 4 months at the Department of Applied Mathematics and Computer Science at the Technical University of Denmark (DTU), in partial fulfilment of the degree Bachelor of Science in Engineering (B.Sc. Eng.) in Software Technology.

This thesis acknowledges the use of Generative AI tools as part of its writing process. The use of these tools followed DTU's ethical principles and Generative AI guidelines. All output assisted by Generative AI tools has been thoroughly reviewed and validated by the authors for accuracy, appropriateness, and compliance with academic integrity requirements. This serves as a disclosure of their use.

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Abstract

Wearables and sensor-integrated footwear represent an active area of research for unobtrusive physiological monitoring. While previous research has established the possibility of capturing cardiac signals from the feet, achieving reasonable precision with simplified hardware remains a technical challenge. This work presents a technical feasibility study for a heart rate detection system based on BCG (ballistocardiogram) measurement using a piezoelectric sensor embedded in a shoe-insole. We demonstrate that signal acquisition is achievable using accessible, off-the-shelf components, lowering the hardware barrier for BCG applications. The system was validated with 13 subject measurements in both seated and standing positions. For subjects in the seated position, the system achieved high precision with an average absolute deviation of 0.91 bpm (beats per minute). A supplementary analysis using independent 20-second windows yielded results within ± 5 bpm for 89.7% of windows, with a mean absolute difference of 3.62 bpm. In contrast, measurements in the standing position exhibited higher variability and lower reliability, mainly due to motion artifacts. Reasonable accuracy was maintained in 5 out of 13 participants, while 20-second windows proved unreliable in this setting. These results suggest that reliable measurements in the standing position remain a hurdle, but low-cost systems are capable of estimating heart rate in controlled seated conditions. While further research is needed, this offers a path towards affordable insole-based wearables.

Acknowledgements

Charalampos (Haris) Orfanidis, Assistant Professor
We wish to thank our supervisor, Haris, for facilitating this thesis and providing input and support throughout the project.

Nicolai Spicher, Associate Professor
We thank Nicolai for his help in guiding the project and his clear-eyed perspective.

Elias Vähälummukka, Friend
We thank Elias for being a wonderful sparring partner and helping the project progress.

Contents

1 Introduction

This chapter provides an introduction to the project, outlining its motivation, scope, and core objectives. We investigate the feasibility of capturing BCG signals through a low-cost, unobtrusive sensor embedded in a standard shoe insole. The following sections define the technical challenges of this methodology using consumer-grade hardware, alongside an overview of our general approach and key results.

1.1 Motivation

Heart rate (HR) is one of the most important indicators for the immediate health of a person. Counting users of wearable devices, one can conservatively assume that more than 100 million people measure their heart rate each day [1].

With the rise of many wearable technologies, it is not unthinkable to have widespread long-term measurements of the heart for a majority of the global population.
With the rise of smart homes, we may see extensive medical monitoring in an ambient and unobtrusive medical setting [2]. This could turn out to be an important feature of future life as there will be a near doubling of old people in the world from the year 2012 to 2050, with the main cause of death currently being cardiovascular disease [3][4].

While wrist-mounted optical sensors currently dominate consumer health tracking, footwear offers a distinct alternative for sensor integration. Smart insoles are already recognized as a viable interface for biomechanical analysis, capable of tracking gait dynamics, posture, and balance [5]. Expanding this existing approach to include cardiovascular metrics represents a logical progression toward comprehensive, multi-modal biometric collection.

This can be achieved using ballistocardiography, which captures the mechanical forces exerted by the heart during the cardiac cycle. Measuring these forces through the sole of the foot eliminates the need for the user to maintain skin contact with a traditional wristband or chest electrodes.

Consequently, establishing that reliable cardiac signals can be extracted using a low-cost piezoelectric sensor on a standard insole offers a practical alternative to current methods.

1.2 Problem definition

While research conducted by Garcia-Limon et al. [6] has demonstrated the ability to extract heart rate signals from an insole-based piezoelectric sensor, their impressive results rely on high-precision laboratory hardware, namely the Measurement Computing Inc.® USB-7202 DAQ. Building upon their robust methodology, the transition from such research prototypes to consumer-ready hardware remains a technical gap.

The primary objective of this project is to evaluate whether reasonable signal accuracy can be maintained when the hardware complexity is significantly reduced. This work focuses on the following goals:

- Replicating the core methodology of the insole-based piezoelectric BCG approach to validate its effectiveness as a heart rate monitoring tool.

- Designing and implementing a signal acquisition pipeline using consumer-grade components and microcontrollers.
- Identifying the limitations that arise under these conditions.

1.3 Approach

To achieve the project goals, our approach was divided into four major phases.

1. Circuit development and piezo integration: We designed and implemented a variant of the circuit in García-Limón et al. [6].
   Integrating our piezoelectric sensor into this circuit required iterative testing and refinement.
2. System prototyping and signal validation:
   After assembling and validating the circuit, we worked on the rest of the signal capture pipeline. A major hurdle in this phase was the amount of noise output by the piezoelectric sensor. This stage was largely iterative, focusing on minimizing noise through various techniques until a recognizable, repeating heartbeat signal could be identified in a controlled, seated setting. We prioritized obtaining a reasonable heart rate estimate in the seated position as a primary milestone, deciding that achieving a reliable heart rate estimate while sitting was a necessary foundation before addressing the motion artifact challenges that are more pronounced in the standing position.
3. Recruiting participants and data collection: We recruited 13 human participants and recorded simultaneous ECG and BCG for each of them, in both the sitting and standing positions.
4. Data analysis: After gathering the data we evaluated the validity and characteristics of the obtained signals.

1.4 Results

We evaluated the system with 13 subjects, recording simultaneous ECG and BCG in both seated and standing positions. Peak to peak detection proved too challenging with our setup, so we estimated heart rate using autocorrelation on each full 2 minute recording instead.

For seated measurements, the average absolute deviation from the ECG reference was 0.91 bpm, with a worst case of 2.52 bpm. Every subject stayed within the AAMI limit of ± 5 bpm [7]. Splitting the same recordings into 20 second windows yielded results within the ± 5 bpm range for 89.7% of windows, with a mean absolute difference of 3.62 bpm.

In the standing position, we were able to estimate heart rate in 5 out of 13 subjects within the ± 5 bpm band using the 2 minute segments. A similar 20 second window analysis proved unreliable with success in only 21.8% of the windows.

This validates that low-cost insole-based sensors can give reasonable HR estimates in controlled, seated conditions while showing some of the challenges associated with estimation in the standing position.

1.5 Thesis structure

The remainder of this thesis is organized as follows. The background chapter introduces the cardiac cycle, ballistocardiography, the electronics used in our circuit, and the signal analysis methods applied to the ECG and BCG data. Related work reviews prior insole based BCG research and provides the context for this project. The design and implementation chapters cover the system's analog design and hardware approach as well as how BCG and ECG measurements were carried out. In the methodology methodology we explain the measurement protocol and how heart rate was estimated from the recorded signals. The experimental findings are presented in results and evaluation and interpreted in discussion, which also considers hardware limitations and trade offs. In future work we outline outline possible next steps, and finally we conclude briefly with an overview of key results.

2 Background

This chapter provides the foundational knowledge necessary to understand the methodology and analysis techniques used in this project. We begin by examining the mechanisms driving cardiac activity. Later, we detail the electronic principles and signal processing strategies used to capture and interpret these biological signals.

2.1 The cardiac cycle

The cardiac cycle refers to the sequence of events during a single heartbeat, consisting of the contraction phase (systole) and the relaxation phase (diastole). This mechanical process is driven by an internal electrical system that ensures the heart's chambers fire in a specific order to pump blood efficiently.

The cycle begins at the sinus node, located in the right atrium. This node acts as the heart's natural pacemaker, generating an electrical impulse that spreads through the atria. In order to give the atria enough time to finish pumping blood into the ventricles, the signal is held briefly at the atrioventricular (AV) Node. Following this delay, the impulse travels rapidly through the ventricles, triggering a powerful contraction.

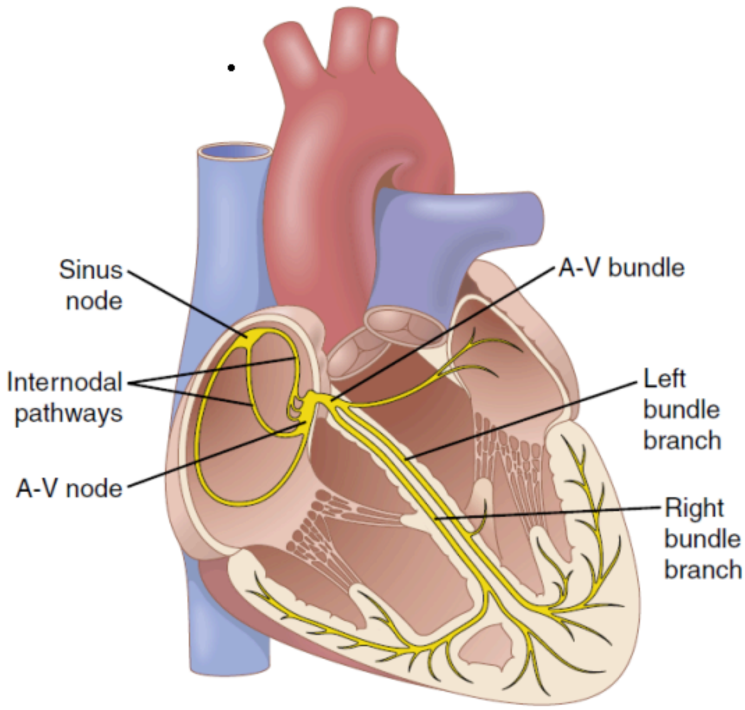

Figure 2.1: Diagram showing the electrical pathways of the heart. Source: [8]

2.2 Electrocardiography

Electrocardiography is the study of the heart's electrical signal over multiple cardiac cycles. In order to capture this signal, you typically have to perform an electrocardiogram (ECG).
The ECG works by measuring the voltage changes across the skin of the subject. This is usually

done by attaching multiple electrodes to the skin. The shape of the ECG waveform reflects the movement of electrical impulses through the heart muscle. The cycle begins with the P wave, representing the initial depolarization spreading from the sinoatrial (SA) node across the atria, followed by a brief electrical pause as the impulse is delayed at the atrioventricular (AV) node, allowing the ventricles time to fill. Next is the QRS complex, which marks the rapid depolarization of the ventricles. Because of their larger muscle mass, the ventricles generate a stronger electrical signal, producing the highest amplitude peaks in the waveform. Finally, the T wave indicates ventricular repolarization, returning the voltage to its baseline and resetting the heart for the next beat.

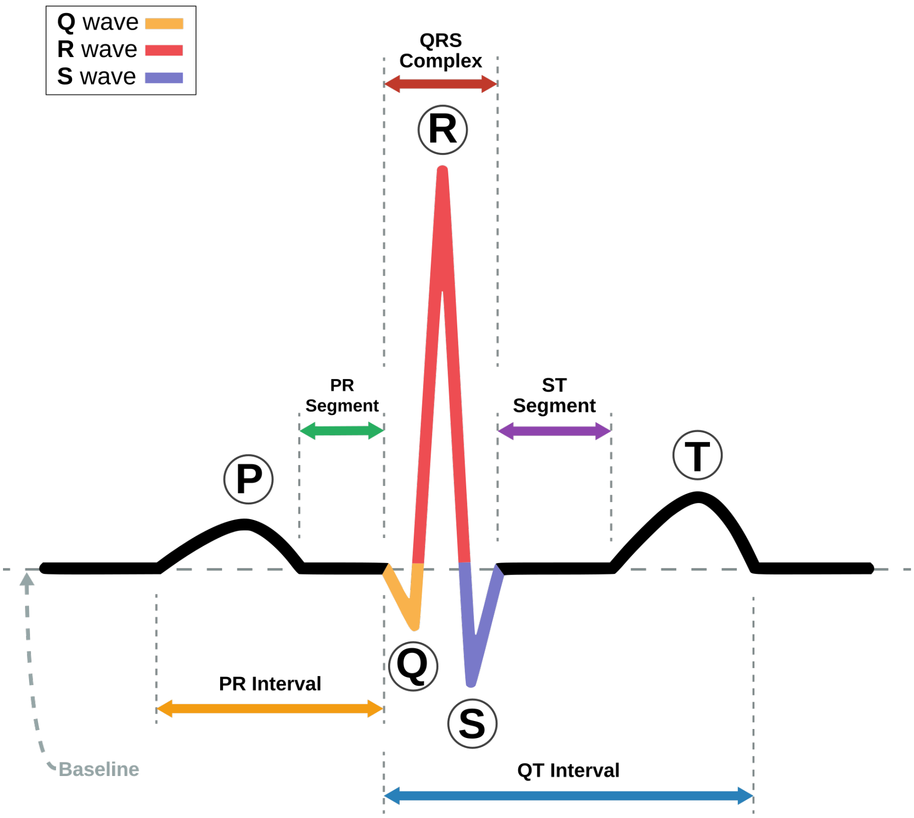

Figure 2.2: Picture of an ECG with the different phases marked. Source: [9]

2.2.1 BPM

A fundamental metric when analyzing any cardiac signal is the frequency of the cardiac cycle, commonly expressed as the heart rate in beats per minute (bpm).
In an ECG signal, the bpm is usually estimated by measuring the time between consecutive R-peaks, which are the most distinct changes in the cycle. This duration is known as the R-R interval (Δt_{R-R}), and the heart rate is determined using the following relation:

$$\text{BPM} = \frac{60}{\Delta t_{R-R}} \tag{2.1}$$

BPM is a key indicator that provides a quick assessment of a subject's immediate clinical status. It is the fundamental metric for identifying arrhythmias, such as tachycardia and bradycardia, and serves as a key indicator of physical form.

2.3 Ballistocardiography

The word ballistocardiography comes from the three Greek words, $\beta\alpha\lambda\lambda\epsilon\iota\nu$ (to throw), $\kappa\alpha\rho\delta\iota\alpha$ (heart) & $\gamma\rho\alpha\phi\epsilon\iota\nu$ (to write), which accurately describes ballistocardiography, as it is the study of how we can write down the heart's throwing or movements.

The early foundation of ballistocardiography was established in 1877 by J.W. Gordon, who first noticed rhythmic movements in a patient standing on a modified weighing scale [10]. Decades later, the methodology was formalized by Isaac Starr, who constructed a specialized, suspended table designed to measure the body's mechanical recoil in response to the ejection of blood from the heart ventricles into the aorta [11]. This early era focused heavily on capturing bigger movements of the body, relying on massive mechanical setups to isolate cardiac signal from environmental noise.

This initial surge of clinical interest lasted until the 1970s and 1980s, when ballistocardiography largely fell out of favor. This decline was mainly caused by the rise of the aforementioned electrocardiogram and echocardiography. The electrocardiogram offered a much clearer and reproducible cardiac signal, while the echocardiogram could provide doctors with quick visual imaging of the heart. Furthermore, early BCG systems suffered from a lack of standardization, required more interpretation of the data from clinicians, and had extreme sensitivity to motion artifacts (subject moving), making them impractical for clinical environments.

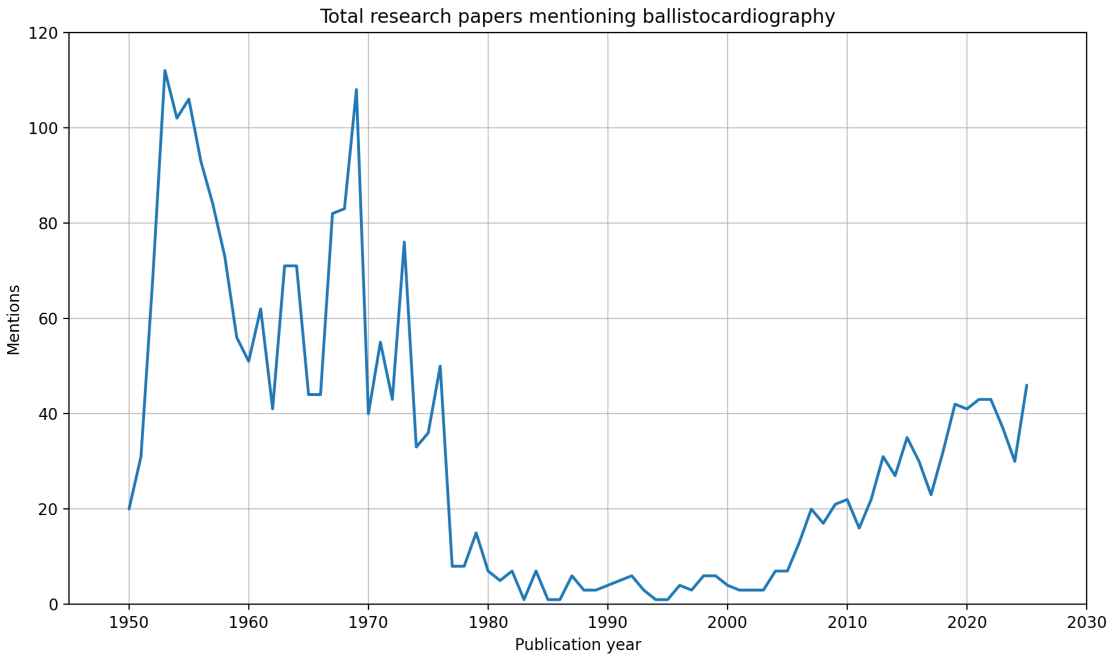

Figure 2.3: Source: [12].

In recent years, the focus in the field of ballistocardiography has shifted from clinical diagnostics to unobtrusive, continuous health monitoring. The advent of advanced embedded systems, improvements in digital signal processing (DSP), and development of highly sensitive microsensors such as piezoelectric sensors have brought renewed interest to the field. By integrating these sensors into everyday objects, contemporary research focuses on capturing body recoil forces passively, avoiding the historical need for bulky systems while using signal processing techniques to isolate the cardiac signal from environmental noise.

2.3.1 BCG methods

There are several different ways to capture a BCG, mostly classified by how the subject is positioned, which sensor is used, and which part of the body touches the sensor.
Since blood flows differently depending on where it is in the cardiovascular system and a BCG measures the physical recoil of the body every time the heart pumps blood, the signal changes based on where the sensor is placed.

Contact areas

To capture a clear BCG signal, the contact surface is typically placed perpendicular to the direction of gravity. This allows the subject's body weight to press naturally against the sensor. This steady pressure establishes a clear baseline, making it much easier for the subtle cardiovascular movements to stand out.

The most common contact areas include:

- **The back:** When a person is lying flat on their back, sensors can be built right into the mattress or the bed frame. This setup catches the movement forces pushing against the back of the torso [13].
- **Gluteal region/thighs:** When someone is sitting in a chair, the cardiovascular forces can be picked up by sensors placed inside or underneath the seat cushion. This setup is very useful for tracking health in office chairs or car seats [14].
- **The Wrist:** Newer smart devices can track BCG signals right at the wrist. Even though the signal is a lot weaker since you are far away from the heart, high-sensitivity sensors can still pick up the tiny vibrations in the arm that happen right after the heart initiates a cardiac cycle [15].

Using the foot as a contact area is an emerging field of research and will be discussed in chapter 3.

Types of Sensors and Equipment

Different kinds of sensors can be used at the contact area to turn the body's recoil forces into a readable signal.

- **Load Cells and Force Sensors:** These sensors measure the tiny changes in total weight and force as the body's center of mass shifts slightly with each heartbeat.
- **Piezoelectric Sensors:** These sensors consist of materials that generate a small voltage whenever they are squeezed or bent. They are thin, flexible and well suited for insertion onto everyday objects. A further discussion of these sensors will follow in section 2.4.1.
- **Accelerometers:** Usually these are tiny, compact chips that measure the literal acceleration of whatever body part they are attached to. They are the main type of sensor used in wristbands and wearables [16].

2.3.2 Decomposition of the BCG

A typical BCG signal consists of a series of waves labeled alphabetically from G to N, though the most clinically relevant portion is the systolic complex, composed of the H, I, J, and K

waves. Each wave corresponds to a specific event within the cardiovascular system during the cardiac cycle.

The cycle begins with the H wave, a subtle upward deflection coinciding with the heart's isometric contraction and the initial headward displacement of blood. This is followed immediately by the I wave, a deep downward trough representing the rapid ejection of blood into the ascending aorta; as the heart accelerates blood toward the head, the rest of the body undergoes an equal and opposite recoil toward the feet.

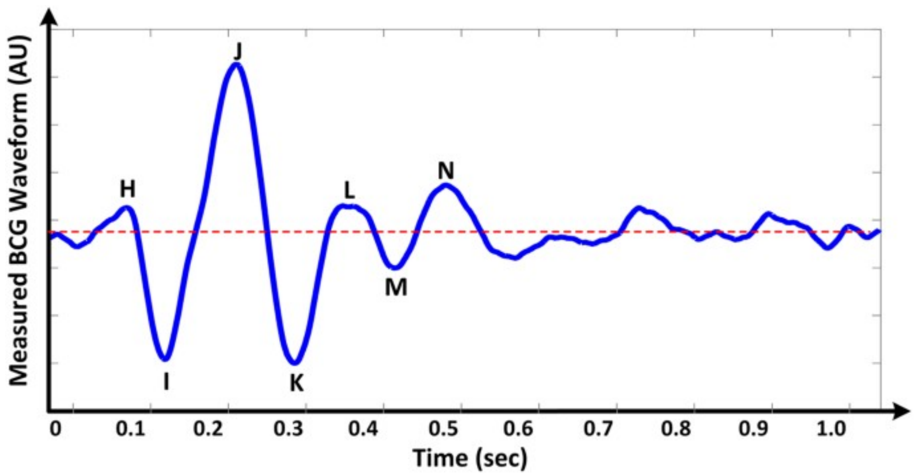

Figure 2.4: The decomposition of a BCG wave. Note the y-axis uses "arbitrary units" [17].

As this pulse of blood rounds the aortic arch and surges into the descending aorta, the direction of momentum shifts, creating the J wave. This is the most prominent upward peak in the signal and serves as the primary mark for identifying individual heartbeats. The cycle then concludes the systolic phase with the K wave, a downward trough caused by the blood decelerating as it meets the peripheral resistance of the lower body. While smaller diastolic waves (L-M-N) may follow as the pressure waves reflect and the heart refills, the primary H-I-J-K complex remains the focus of most analyses.

Because the R-peak from the ECG precedes the mechanical ejection of blood, there is an inherent delay between R-peaks and J-peaks, the length of which varies in part depending on the distance from the heart to the specific BCG contact area.

2.4 Electronics

2.4.1 Piezoelectric sensors

The word piezoelectricity is derived from the Greek word *piezein*, meaning to squeeze or press. It describes a physical phenomenon where certain crystalline materials and ceramics generate an electric potential in response to applied mechanical stress.

The underlying principle of a piezoelectric sensor is the displacement of ions within the material's crystal lattice. Mechanical deformation shifts the balance of positive and negative charges, creating an internal electric dipole moment. This results in the accumulation of opposite charges on the sensor's surfaces, which can be measured as a voltage (V). Below is a picture illustrating the piezoelectric effect on a quartz (SiO_2) crystal:

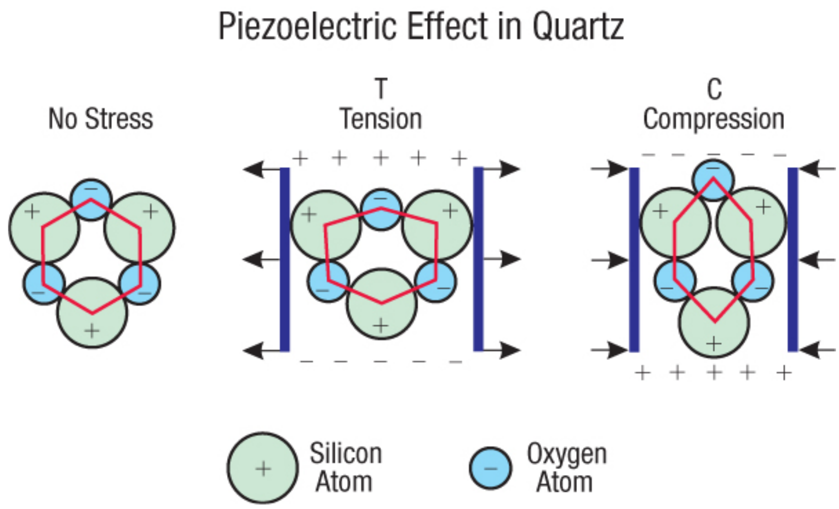

Figure 2.5: Source: [18]

Several materials can be used for piezoelectric purposes, depending on the mechanical requirements of the application, including quartz (SiO_2), Lead Zirconate Titanate (PZT), and Polyvinylidene Fluoride (PVDF).
The latter material, PVDF, is a ferroelectric polymer that offers a distinct alternative to rigid crystals and ceramics due to its high mechanical flexibility and impact resistance. Following the discovery of its piezoelectric potential by Heiji Kawai in 1969, it has become a standard for unobtrusive sensing applications [19][20].

While piezoelectric sensors elegantly capture cardiac forces, their raw output is a noisy, high-impedance signal that cannot easily be read directly. Therefore, a dedicated analog front-end is required to filter and amplify the signal before digitization. To understand why this is the case, it is helpful to look at an electrical model of a piezoelectric sensor.

Electrically, the sensor functions as an AC charge source represented as the rate of change of charge, $\frac{dq}{dt}$, connected in parallel with its own internal capacitance (C_o) and internal leakage resistance (R_o).

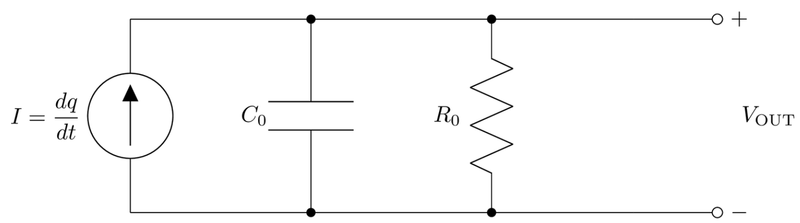

Figure 2.6: Electrical model of a piezoelectric sensor

Because the internal resistance R_o is rather large, the sensor exhibits a high source impedance. Furthermore, the internal capacitance C_o is inherently unpredictable. It fluctuates based on temperature and the slightest physical shifts or deformations of the sensor. If read directly, the fluctuating C_o would cause the total voltage to shift erratically and seemingly at random. To capture a usable signal, the raw sensor output therefore requires dedicated analog filtering and amplification, which we address in the following sections.

2.4.2 Electronic Devices

Only a few types of electronic device were used in this project. Keeping the component count low made the circuit simple to understand, and hopefully simple to follow.

Capacitors

Figure 2.7: The ANSI symbol of a capacitor[21]

In its simplest form, a capacitor is two metal plates facing each other, separated by an insulating material. Their defining property is the ability to store charge Q, given by

$$Q = CV \tag{2.2}$$

Capacitance also depends on the plate area and spacing, proportional to area and inversely proportional to spacing, though this is outside the scope of the thesis.
Capacitors behave differently under direct current (DC) and alternating current (AC). Under DC, a capacitor charges up and, once full, blocks any further current, much like a bucket that can hold no more water. Under AC, current continues to pass. The reason becomes clear once we note that current is the derivative of charge Q. Differentiating Equation 2.2 gives

$$I = C\frac{dV}{dt} \tag{2.3}$$

This shows that the current through a capacitor is not proportional to the voltage, but to its rate of change.
This makes capacitors useful for removing static DC components from a signal.

2.4.3 Impedance

The theory so far has dealt mainly with DC signals, which are relatively simple. To analyse AC behaviour we need one further concept: **impedance**, the total opposition of a circuit to current.

It has two parts, combined into a single complex number:

$$Z = \underbrace{R}_{Resistance} + \underbrace{jX}_{Reactance} \tag{2.4}$$

Representing it this way makes combining signals across components straightforward. This can be seen in the voltage divider.

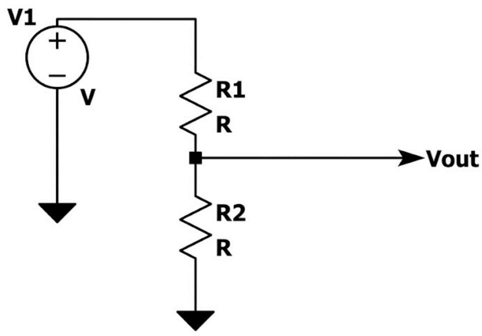

Figure 2.8: A schematic of a voltage divider

The voltage is divided between V_{out} and R_2, according to

$$V_{out} = V_{in}\frac{Z_2}{Z_1 + Z_2} \tag{2.5}$$

A high impedance on the second component leaves a larger signal at V_{out}. For resistors this is straightforward, since the impedance is simply the real part of Equation 2.4.

2.4.4 Amplifiers

An amplifier is a device that amplifies an electrical signal, usually AC, drawing on a DC power supply. This amplification is called gain.

$$V_{out} = A \times (V(+) - V(-)) \tag{2.6}$$

Where $V(+)$ & $V(-)$ are the two input ports and A is the gain of the amplifier.
The output is the product of this gain and the difference between the two input nodes:

$$V_{\text{diff}} = V(+) - V(-)$$

Amplifiers have other uses. One is to create a virtual ground, by connecting the non-inverting input $V(+)$ to ground and feeding the output back to the inverting input$V(-)$. The voltage at the node then sits at approximately $0\,\text{V}$.

This works because negative feedback is self-correcting. Suppose $V(-)$ drifts slightly negative. Then $V(+)-V(-)$ is no longer 0, and in equation 2.6 that difference gets multiplied by A, so the output swings hard positive. But that output is fed straight back into the inverting input, dragging $V(-)$ back up toward 0. The same happens in reverse: a positive $V(-)$ pushes the output negative, pulling $V(-)$ back down. This is summed up by

$$V(+) - V(-) = \frac{V_{out}}{A} \tag{2.7}$$

Since V_{out} can only swing within the supply rails while A is on the order of 10^6, the input difference is a few micro volts at most, negligible. Hence $V(-) \approx V(+) = 0$, and the node behaves as a virtual ground.

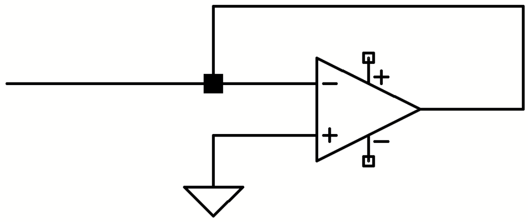

Figure 2.9: A virtual ground created between the node and the amplifier

Amplifiers can also act as buffers between two parts of a circuit.
Placing an amplifier between two filters is advantageous because it presents effectively infinite input impedance and virtually no output impedance.
Starting from Equation 2.5 as part of a filter, suppose an additional load Z_L is added in parallel with Z_2. The expression becomes

$$\frac{Z_2 Z_L}{Z_2 + Z_L}$$

If $Z_L \gg Z_2$, the other term in the denominator becomes negligibly small,

$$\frac{Z_2 \cdot Z_L}{Z_L} = Z_2$$

keeping the signal as before.
The output impedance is likewise negligible, as seen in

$$V_{out} = V_{source} \cdot \frac{Z_L}{Z_{out} + Z_L}$$

When $Z_{out} \ll Z_L$, this reduces to V_{source}, giving a clean signal to the next stage.

A itself can vary widely with temperature, frequency, and amplitude [22], so it is preferable to reduce the circuit's dependence on it. This is done by introducing feedback.

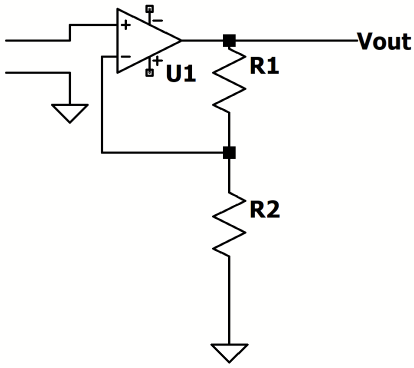

Figure 2.10: Feedback on a non-inverting amplifier

Part of the output is fed back through a voltage divider into the $V(-)$ input. This gives

$$V_{\text{diff}} = V_{\text{in}} - \frac{R_2}{R_1 + R_2} V_{\text{out}} \tag{2.8}$$

Writing the return fraction simply as B, the output becomes

$$A(V_{\text{in}} - BV_{\text{out}}) = V_{\text{out}}$$

which can also be written as

$$V_{out} = \frac{A}{1 + AB} V_{\text{in}}$$

so the gain is

$$G = \frac{A}{1 + AB} \tag{2.9}$$

[22, p. 250]

2.4.5 Filters

Passive filters

Adding a capacitor to the voltage divider, as shown here,

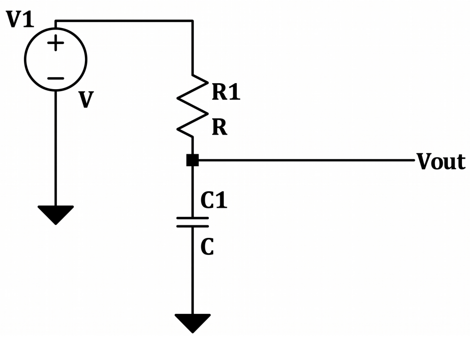

Figure 2.11: A low-pass filter

gives

$$f_c = \frac{1}{2\pi RC} \tag{2.10}$$

This result is the **cutoff frequency**, the point beyond which the signal is no longer passed. It is also called the -3 dB point, since here the output power is roughly halved and the voltage falls to about 71% of its input. The roll off is gradual, but this point is taken by convention as the cutoff. Equation 2.10 therefore gives the cutoff frequency for a given resistor and capacitor.[22, p. 49] Beyond the cutoff, the response falls off at a fixed rate, described per octave. An octave is always a 2:1 frequency ratio. So increasing a frequency by one octave is a doubling of that frequency. For a first-order filter, the output drops by a further 6 dB per octave, a factor of two in voltage.

2.4.6 Sallen-Key filter

A steeper rolloff than 6 dB/octave is often needed. The ideal filter would look like the following

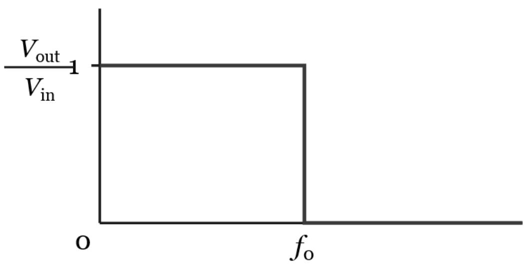

Figure 2.12: An ideal filter

This is achieved by increasing the order of the filter, which corresponds to using more capacitors.
Simply chaining passive filters together is not enough. The result stays well short of the ideal in Figure 2.12[22, p. 391].
The problem with a chain of RC filters is that each stage loads the one before it. This means that it draws current from the previous stage, distorting the cutoff frequency; a buffer would solve this issue.
It also does not produce a sharp drop-off. Even if the stages were buffered so they did not load each other, each section still contributes its own -3 dB at the corner frequency, so n stages together give $-3n$ dB at that point. To obtain a sharper response with a flat passband, the resistor and capacitor values must interact in a way that a simple passive chain cannot achieve on its own.
This interaction is made possible by introducing feedback with an operational amplifier. This is precisely what Sallen-Key does: the op-amp feeds part of the output back into the *RC* network, so that the stages no longer behave independently.
By choosing the resistor and capacitor values, the designer can shape the response of a single second-order stage, setting both its cutoff frequency and how sharply it peaks near the cutoff, and thereby approximate the ideal far better than a passive chain of the same size.

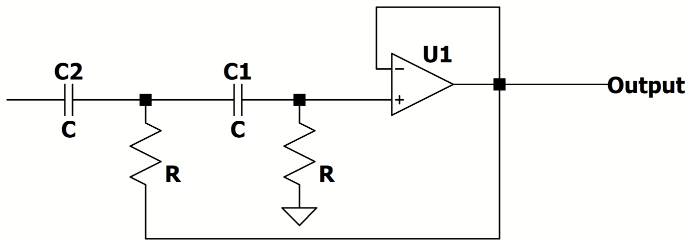

Figure 2.14: Sallen-Key highpass filter

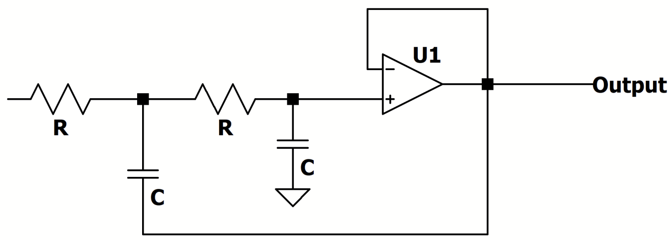

Figure 2.13: Sallen Key low pass filter

Cutoff and quality factor

A second-order stage is described by two numbers. The first is the cutoff frequency f_0, which sets where the corner sits. The second is the quality factor Q, which sets what the response does at that corner before it rolls off. In practice Q describes how much the stage peaks near the cutoff: a low Q gives a gentle, well-damped corner with no peak, while a high Q gives a sharper corner with a peak just before the rolloff.[22, p. 53]

Butterworth response

The peaking behaviour is what separates the standard filter families. The **Butterworth** filter chooses the Q that gives the flattest possible passband without any peak, at the cost of a slightly softer corner than a peakier design would give. This is why it is called maximally flat, and it is the response targeted in this project.

There is no single "Butterworth Q", it depends on the order of the filter. A second-order Butterworth uses a single stage at $Q = 0.707$. A higher even order is built by cascading second-order stages, each with its own Q. An odd order adds a first-order real pole (a plain RC, with no Q) to the cascade[23].

This project uses a third-order Butterworth. From the filter tables, this decomposes into one real pole and one second-order stage with $Q = 1$[23]. The real pole sets one corner, and the

Sallen-Key stage realises the $Q = 1$ section. The component values for that stage are chosen to give $Q = 1$ while keeping the desired f_0, as described in Section 4.1.

2.5 Signal analysis

In this section we lay out different signal analysis techniques for heart rate estimation, namely the Pan-Tompkins algorithm for the ECG and autocorrelation, which will be used for BCG analysis.

2.5.1 Signal Energy

Before analyzing a signal's frequency, it is often necessary to determine its overall magnitude or strength. In digital signal processing, we therefore often use the "energy" of a discrete finite sequence $x[n]$ of length N, defined as the sum of its squared samples:

$$E = \sum_{n=0}^{N-1} x[n]^2 \tag{2.11}$$

Squaring the signal transforms it into only non-negative values, which prevents troughs and peaks from being cancelled out by one another.

2.5.2 The Pan-Tompkins Algorithm

After capturing an ECG, there are multiple ways to digitally process the data to determine the heart rate.

While several approaches exist, the Pan-Tompkins algorithm is a long-standing, reliable method.

The Pan-Tompkins algorithm starts by applying a bandpass filter to remove noise, followed by a derivative stage to isolate the steep slopes of ventricular depolarization. Mathematically, this emphasizes the R-peak where $\frac{dV}{dt}$ is at its maximum.

The filtered derivative is squared to rectify the signal and amplify the R-peaks:

$$y[n] = x[n]^2 \tag{2.12}$$

A Moving Window Integral (MWI) then aggregates this energy over a rolling timeframe. Taking the squared signal $y[n]$ as its input, the MWI outputs $z[n]$:

$$z[n] = \frac{1}{N} \sum_{i=0}^{N-1} y[n-i] \tag{2.13}$$

where the window width N is chosen to encompass the QRS complex, preventing double-triggering on jagged peaks.

To account for amplitude fluctuations, a dynamic threshold is maintained based on the running signal level (THR_{SIG}) and noise level ($\text{THR}_{\text{NOISE}}$):

$$\text{Threshold} = \text{THR}_{\text{NOISE}} + 0.25(\text{THR}_{\text{SIG}} - \text{THR}_{\text{NOISE}}) \tag{2.14}$$

If no peak is detected within 166% of the average R-R interval, a backsearch is initiated using a lower threshold to capture potentially missed beats [24].

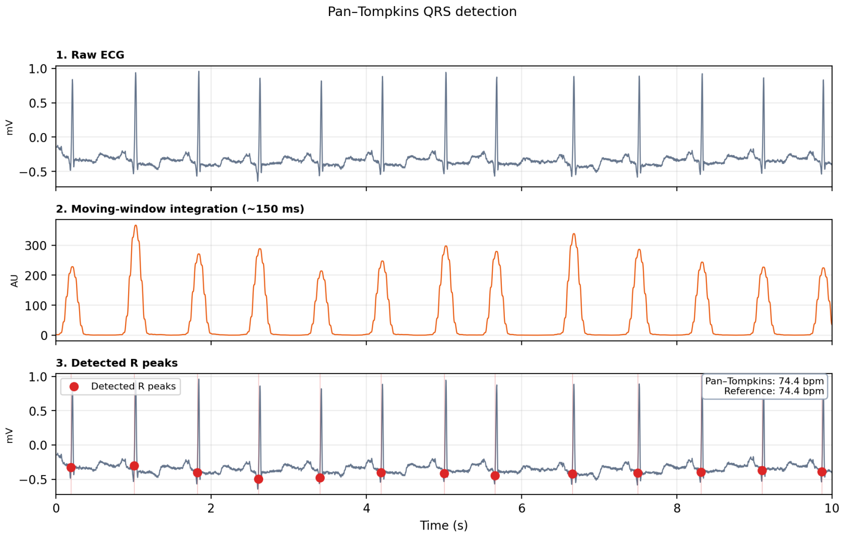

Figure 2.15: An illustration of an ECG data analysis pipeline on a generic ECG measurement [25].

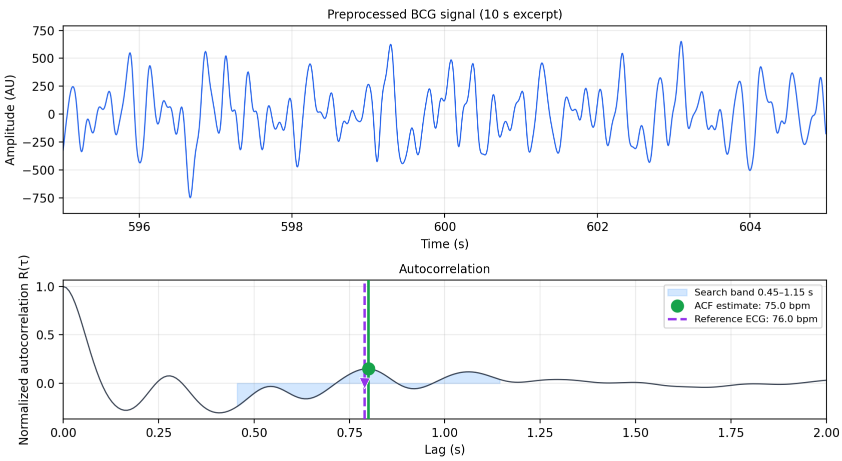

Figure 2.16: Autocorrelation function applied to a generic preprocessed BCG measurement [26].

2.5.3 Autocorrelation

Autocorrelation identifies a signal's periodicity by measuring how well its energy aligns with a shifted version of itself. For a discrete finite sequence $\{x[n]\}_{n=0}^{N-1}$, an empirical autocorrelation at an integer lag k is formed by correlating the signal $x[n]$ with $x[n-k]$ over the overlap region. This is mathematically expressed as:

$$R_{xx}[k] = \sum_{n=k}^{N-1} x[n] \cdot x[n-k] \tag{2.15}$$

At a lag of zero ($k = 0$), the signal perfectly overlaps with itself, making the expression equal to the total signal energy. As the lag k increases, large positive values indicate that the energy aligns with its own previous cycles. When the raw signal is noisy, autocorrelation still reinforces underlying rhythms, producing useful local maxima.

Strengths and weaknesses
Autocorrelation has been shown to be a useful estimation technique for determining BPM in a BCG measurement, particularly in high-noise environments [27]. Because noise is largely random, its energy does not correlate with itself over long shifts and is suppressed after summation. Meanwhile, the periodic heartbeat signal reinforces itself at every lag k that equals a multiple of the heart rate period.

However, the method possesses significant weaknesses regarding individual peaks, since it aggregates the signal's energy across a finite observation window to find a global average. Therefore, autocorrelation is a tool for robust frequency estimation rather than a method for detailed peak analysis.

3 Related Work

In this chapter, we lay out the current landscape of biometric heart rate monitoring using ballistocardiography. The field has seen a resurgence in the last decade and shows promise as a practical, unobtrusive tool. We start by examining foot-based BCG methods, specifically how recent research has measured cardiac forces from the sole.
We then give an overview of the signal analysis methods used in this context. Finally, we lay out our contribution by identifying the technical hurdles that remain and the specific gap this thesis aims to address.

3.1 Foot-based BCG

The foot provides an interesting interface for unobtrusive cardiovascular monitoring. When a person is seated or standing, the forces stemming from the heart displace the body's center of mass, generating a corresponding reaction force at the surface of the foot. Therefore, the sole acts as a natural mechanical interface where impulses can be captured passively, eliminating the need for direct skin-to-electrode contact as in ECGs. However, translating this to wearables requires isolating these small forces from environmental and motion noise.

The technical foundation for our work was established by García-Limón et al. [6], who demonstrated the ability to estimate heart rate using a piezoelectric sensor embedded in a shoe insole. When conducting measurements, they placed the sensor in the heel area of the foot rather than the metatarsal area, which has been the standard focus of traditional pressure sensing insoles aimed at gait analysis and diabetic foot monitoring [5].

Their system reliably identified J-peaks, and therefore also heart rate in both the seated and standing position. Building on this in another paper they were able to show that insole BCG can be used to estimate heart rate variability (HRV) [28].

However, their experimental setup used expensive laboratory data acquisition hardware (Measurement Computing Inc. USB-7202 DAQ). This leaves an open question regarding whether such performance can be replicated using low-cost, consumer-grade components.

3.1.1 Multimodal sensor approaches

While standalone BCG provides a viable method for heart rate estimation in static positions, mechanical signals are susceptible to motion artifacts.

To overcome the limitations of relying solely on mechanical displacement, recent research has explored multimodal sensor approaches. A prime example is the subsequent work by García-Limón et al. [29], which expanded their insole approach to capture both BCG and Impedance Plethysmography (IPG) signals simultaneously. By measuring localized blood volume changes via IPG alongside whole-body mechanical recoil via the piezoelectric BCG sensor inside the same shoe, they demonstrated that combining electrical and piezo-based sensors significantly improves the robustness of heartbeat extraction, particularly in the presence of noise.

Beyond IPG, other approaches have explored combining BCG with Photoplethysmography (PPG) [30]. By combining the mechanical BCG signal with PPG data, systems can cross-reference the timing of cardiac blood ejection with the arrival of the pulse at extremities. This mitigates the challenge of motion artifacts and allows for the extraction of secondary metrics like Pulse Transit Time (PTT).

3.1.2 Noise Filtering and Motion Artifact Rejection

One major hurdle for HR estimation from BCG signals is the presence of motion artifacts, which can easily overwhelm the micro-vibrations of the heart. To address this, recent work has performed dedicated artifact detection.

García-Limón et al. [6] implemented a robust motion artifact rejection algorithm. In addition to using standard bandpass filters, their system evaluated the signal's energy over short, rolling windows. When the mechanical energy exceeded a dynamic threshold, that specific segment was explicitly flagged and discarded. By selectively cutting out these corrupted intervals, their further analysis could proceed with clean data.

3.1.3 HR estimation from BCG signals

Extracting a reliable heart rate from BCG signal is a known challenge. To avoid the pitfalls of simple time domain peak detection, researchers typically look to the frequency domain.

García-Limón et al. [6] demonstrated beat to beat detection using the Continuous Wavelet Transform (CWT). Other approaches focus on finding the average heart rate over a specific time window rather than detecting individual beats. For example, Elnaggar et al. [27] used autocorrelation based methods which effectively suppressed non periodic noise. Similarly, Kortelainen and Virkkala [13] employed Fast Fourier Transform (FFT) averaging to isolate the dominant heart rate frequency from bed mattress sensors.

Ultimately, whether utilizing CWT, ACF, or FFT, the success of these HR estimation algorithms relies heavily on robust data and pre-processing.

3.2 Contribution

The contribution of this thesis is a technical feasibility study of low cost insole-based BCG heart-rate estimation. Building on the general approach demonstrated by García-Limón et al. [6], we implemented a simpler prototype using consumer grade components and tested it against simultaneous ECG measurements.

The main contribution is showing that useful heart rate estimates can be obtained in the seated position without specialized laboratory equipment. This supports the idea that insole-based BCG can be made more accessible and less dependent on expensive hardware. At the same time, the standing measurements show a clear limitation. The system we present had mixed results when estimating heart rate in this setting. This shows the promises of the insole based BCG approach while highlighting current challenges.

4 Design

This chapter details the analog circuit and hardware designed to capture the ballistocardiography (BCG) signal. The charge amplifier and middle stage are built around MCP6002 op-amps, with an OP270 in the final Butterworth stage. This differs from García-Limón et al. [6], who used an AD8641 and an OP207 for the first two stages.

4.1 Circuit design

Our circuit design was heavily inspired by the circuit from García-Límon et al. [6]. The design is elegant in its way of utilizing both high pass & low pass filters to gain the proper signal through.
The end goal is to remove as much unnecessary noise as possible and remove all signals that didn't come into the 0.5-20 Hz Window.

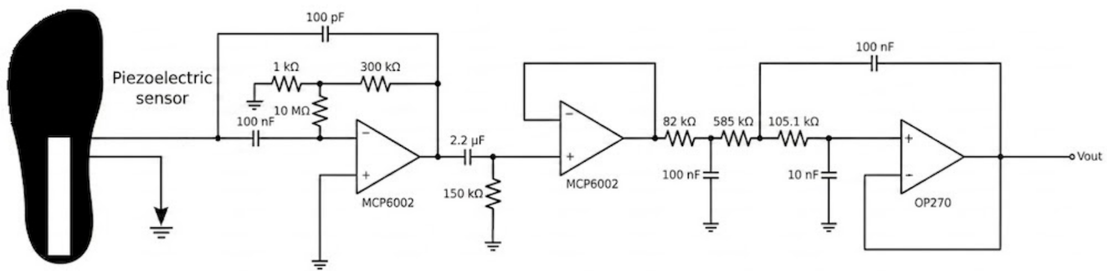

Figure 4.1: The full Circuit with the piezoelectric sensor

The piezoelectric sensor's output is generally unpredictable and changes according to the slightest shifts in force being applied to it, meaning the total voltage would shift seemingly randomly. Therefore the first section of the circuit used a charge amplifier.

4.1.1 Charge Amplifier

The first section consists of a charge amplifier and a T-shaped high pass filter.

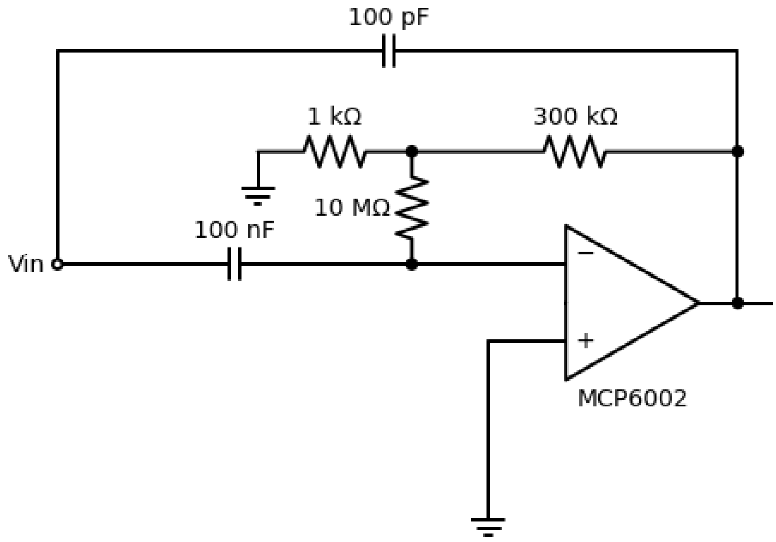

Figure 4.2: Section 1 of the circuit

The Piezo electric sensor is in effect a *charge generator*. Because the signal must be in voltage, we use the following formula. Which states that the current in a capacitor is equal to the capacitance times the rate of change of voltage. And we remember that current is just charge over time, see equation 2.3.
By integrating for time you get

$$Q = C_f V_{out} \tag{4.1}$$

$$V_{out} = -\frac{Q}{C_f} \tag{4.2}$$

Thus it is possible to change the current into voltage.

Virtual Ground
As previously mentioned, amplifiers with negative feedback will always attempt to have the two inputs equal each other.
Since one of the inputs $V(+)$ is in ground (0 volts). It means that it ensures that the input at $V(-)$ is also 0 V. This means that all the charge flows into the feedback network instead into the capacitor in the sensor.
The current can only move through the feedback network, because there is no voltage in the bottom part of the piezo electric sensor because of the ground. Meanwhile because of the op-amp has created virtual ground there is no ground on the other side either.
Since there is no voltage, there can't be any capacitance on the piezo electric sensors capacitor (denoted by C_0) therefore it can only move through the feedback. Here the current is made into voltage as previously mentioned.

This makes the output only dependent on charge, and not the piezo electric sensor's capacitance which varies with temperature etc. The resistors $1k\ \Omega, 10\ M\Omega$ & $300k\ \Omega$ act together with the 100 pF capacitor as a high pass filter. The high pass filter is made up of the resistors in a T-network.

$$300k\ \Omega + 10M\ \Omega + \frac{300k\ \Omega \cdot 10M\ \Omega}{1k\ \Omega} \approx 3G\ \Omega$$

Putting that into equation 2.10, we gain the following

$$\frac{1}{2\pi \cdot 3G\Omega \cdot 100pF} \approx 0.5\text{Hz}$$

4.1.2 First order high pass filter

The capacitor in the beginning and the resistor create a high pass filter see. 2.4.5 This reduces all signals below the frequency, by 6 db.

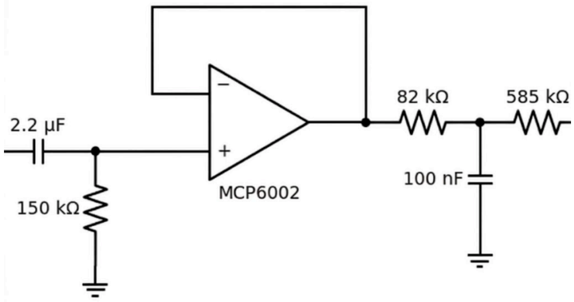

Figure 4.3: Section 2 of the circuit

$$\frac{1}{(2\pi \cdot 150\ k\Omega \cdot 2.2\ \mu F)} \approx 0.48\ Hz \tag{4.3}$$

any signal below 0.48 Hz will be greatly limited.
The feedback stands to act as a "buffer" between the two parts of the circuit. By having the output be equal to the input,we gain a gain of 1. See equation 2.6 So there is no gain, but its acts with a massive impedance in the beginning. Thus it doesn't act as there is more connected to the circuit. Therefore it keeps the cutoff frequency as it is. Without it the $2.2\ \mu F$ & $150\ k\Omega$ high pass filter would see the $82\ k\Omega$ in parallel with the $150\ k\Omega$ thus changing the filter to

$$\frac{150\ k\Omega \cdot 82\ k\Omega}{150\ k\Omega + 82\ k\Omega} \approx 53\ k\Omega \tag{4.4}$$

$$f = \frac{1}{2\pi\ 53k\Omega\ 2.2\mu F} = 1.37Hz \tag{4.5}$$

So amplifier is needed for dividing up the circuit and preventing impedance along the circuit.

4.1.3 Butterworth filter

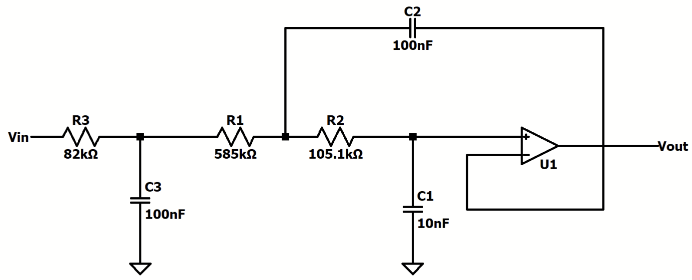

Figure 4.4: A third order butterworth filter

The circuit in Figure 4.4 acts as the final part of the circuit. In it is a third-order Butterworth filter, made up of a first-order RC low-pass filter followed by a second-order Sallen-Key stage. This can be seen by the three capacitors. Butterworth filters are uniquely good at attenuating the signal past the cutoff frequency while keeping a flat passband.

The first-order real pole is set by the 82 kΩ resistor and the 100 nF capacitor to ground:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(82\,\mathrm{k\Omega})(100\,\mathrm{nF})} \approx 19.4\,\mathrm{Hz} \tag{4.6}$$

The second-order section is the Sallen-Key stage built around the OP270, with $R_1 = 585\,\mathrm{k\Omega}$, $R_2 = 105.1\,\mathrm{k\Omega}$, a grounded capacitor $C_1 = 10\,\mathrm{nF}$ at the non-inverting input, and a feedback capacitor $C_2 = 100\,\mathrm{nF}$. The op-amp is wired as a unity-gain follower ($K = 1$), so the general Sallen-Key expressions [23] reduce to

$$f_0 = \frac{1}{2\pi\sqrt{R_1R_2C_1C_2}} \approx 20.3\,\mathrm{Hz} \tag{4.7}$$

$$Q = \frac{\sqrt{R_1R_2C_1C_2}}{R_1C_1 + R_2C_1} \approx 1.14 \tag{4.8}$$

The $R_1C_2(1-K)$ term in the denominator vanishes at unity gain, leaving Q set purely by the resistors and the grounded capacitor.

A true third-order Butterworth requires its second-order stage to have $Q = 1$ [23]. The realized value of $Q \approx 1.14$ is therefore about 0.14 above target. The consequence is a slight peak near the corner frequency, so the passband is marginally less flat than an ideal Butterworth and leans very slightly toward a Chebyshev response. This deviation stems from the chosen component values; hitting $Q = 1$ exactly at $f_c = 20\,\text{Hz}$ with these capacitors would instead require $R_1 \approx 706\,\text{k}\Omega$ and $R_2 \approx 90\,\text{k}\Omega$.

The Butterworth stage also serves as an anti-aliasing filter. The signal is digitized at a sampling rate of 200 Hz, so by the Nyquist criterion any content above 100 Hz (half the sampling rate) would otherwise fold back into the band of interest and be indistinguishable from genuine signal once sampled[22, p. 395].

By attenuating frequencies above its $\sim$20 Hz corner, well below the 100 Hz Nyquist frequency, the filter band-limits the signal before it reaches the ADC, leaving a wide margin against aliasing.

5 Implementation

This chapter outlines the physical layout of our system. We detail the circuit board layout followed by the signal capture pipeline of both the BCG and reference ECG.

5.1 Circuit board layout

The circuit was firstly developed on a breadboard, namely because of its relative ease to set up. But also because it was easy to reorganize the external components in the case of any errors or changes.
We were given two MCP6002 and one OP270 amplifiers. The two MCP6002 were used for the charge amplifier and the middle stage, while the OP270 was used for the final Butterworth stage. This differs from the circuit by García-Limón et al. [6], which used an AD8641 at the charge amplifier and an OP207 at the middle stage. We kept the OP270 in the final stage, but substituted the cheaper, more accessible MCP6002 for the first two, in keeping with the low-cost goal of the project. The consequences of this substitution are discussed further in the discussion chapter.

The circuit was powered separately from the Arduino. The analog circuit was supplied with 4.9 V across the supply rails of the breadboard, which also carried the ground, keeping us safely below the 5 V threshold to avoid damaging any components. The Arduino had its own power supply at 7 V, satisfying its voltage requirements. Keeping the two supplies separate also helped avoid the analog circuit and the Arduino interfering with each other.

One consequence of the breadboard setup was that the piezoelectric sensor was not soldered, but connected to the circuit through the breadboard contacts like everything else. This made it easy to reposition or swap the sensor, but it also meant the connection was fragile and sensitive to movement, which became a problem during the standing measurements.

Since the components were only held in place by the breadboard's contacts, the layout was kept as compact as possible to keep the wires short and reduce the amount of outside noise picked up. This was especially important at the high-impedance front end, where long, exposed wires would have acted as antennas for the 50 Hz interference present in the lab.
The full breadboard wiring is shown in section A.2 of the appendix.

5.2 BCG setup

For the BCG measurement the following major components were used:

- 0.5 mm thick DT4-028K piezoelectric sensor
- Breadboard with our electrical circuit
- Arduino UNO R4 Minima microcontroller
- Laptop for data capture and visualization
- Power supply for the arduino
- Agilent DC Power Supply for the circuit

To ensure stable operation, the power distribution was carefully managed. The Arduino UNO was powered at 7V, satisfying its voltage requirements, while the analog circuit was supplied

with 4.9V, ensuring that we remained below the 5V threshold to prevent potential component damage. The power supplies were plugged into the same power strip and the laptop was never charging during measurements. The sensor signal was routed through the breadboard circuit for initial filtering and amplification before being digitized by the Arduino's ADC and transmitted to the laptop via USB.
The Arduino UNO R4 was chosen due to its processing power and its ability to output true analog signals (DAC) which proved useful when validating the circuit.

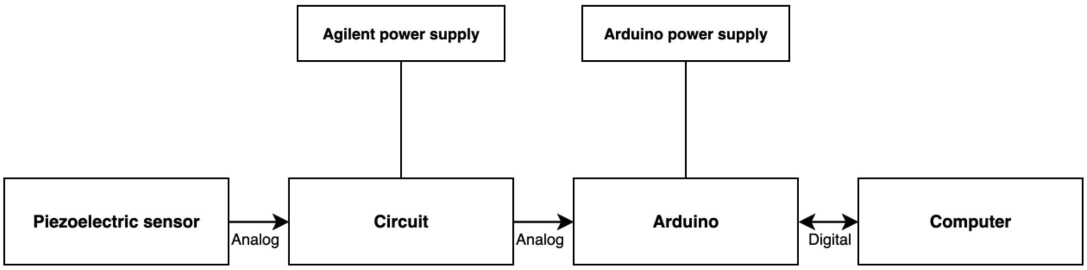

Figure 5.1: Diagram illustrating the data flow in the BCG setup.

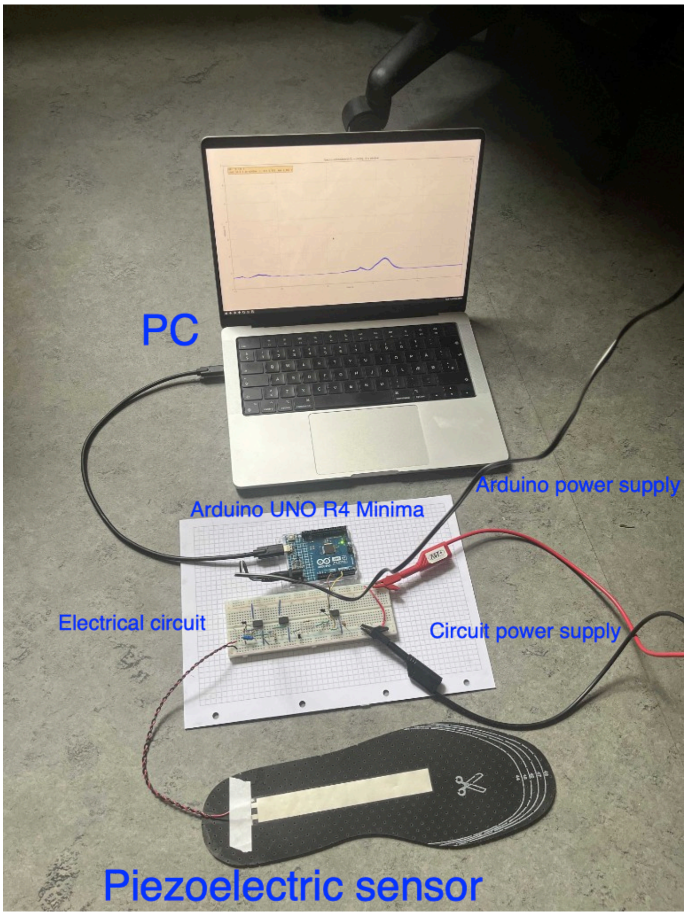

Figure 5.2: Picture of the BCG setup

5.3 ECG setup

For the ECG measurement the following components were used:

- SparkFun Single Lead Heart Rate Monitor (AD8232)
- Arduino Uno Microcontroller
- Laptop for data capture and visualization
- Three Disposable Ag/AgCl Surface Electrodes

The SparkFun device would capture the heart signal, send it to the Arduino through a breadboard connection, which would send it to the laptop through a USB connection after which the data would be saved and plotted in real-time to ensure a reliable signal was being captured.

The length of an ECG lead wire was 61 cm, therefore it was important to have the ECG setup at a height where both the sitting and standing measurements could be taken.
For this purpose a wooden rolling cart was used. As a side benefit this avoided electrical interference between the BCG and ECG equipment, which was observed when the components were on the same surface.

The 3 electrodes were placed below the left and right collarbone and below the subject's rib cage on the right, adhering to Einthoven's triangle [8].

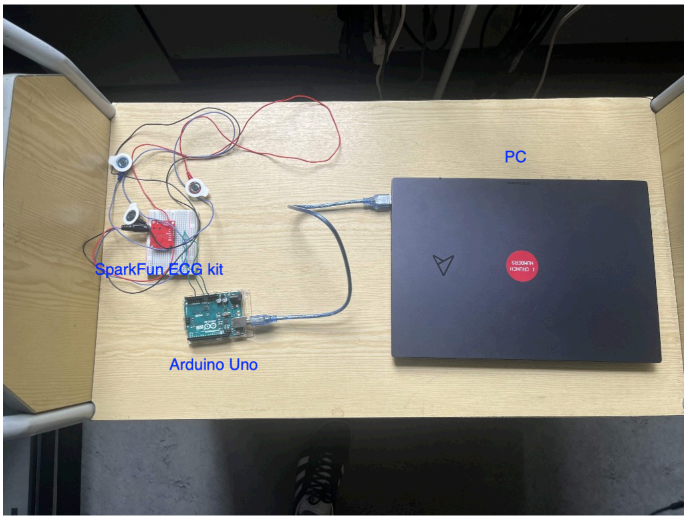

Figure 5.3: Picture of the ECG setup

6 Methodology

This chapter outlines the experiment specifics and data processing pipeline used to evaluate the insole-based BCG system. We first describe the subject characteristics and the measurement protocol. We then detail the digital signal processing, and estimation techniques used to extract heart rate values.

6.1 Data collection

Prior to participating in this study all subjects were given a written consent form. A total of 13 healthy subjects were recruited. The participants were:

- All male
- Age: 23.7 ± 4.7 years
- Height: 185.9 ± 5.6 cm
- Weight: 83.9 ± 10.2 kg

The experiment room was a standard electronics lab at DTU (ESE lab) with a temperature around $19°$C.

6.1.1 Measurement procedure

For each measurement, the subject was instructed to sit comfortably on an office chair with their knee at a $90°$ angle and their heel at the wire end of the piezoelectric sensor. Disposable electrodes were attached to the subject's upper body, followed by a brief visual "sanity check" to ensure a reasonable signal was being picked up.

Recording sessions were then scheduled by a terminal script on each computer, that started logging and visualization on both the ECG and BCG systems at a specific time of day. Each recording lasted 3 minutes. The first minute served as a buffer, allowing the authors to provide guidance on posture to ensure the subject attained a position that provided a clear signal. After the seated measurement was completed, the subject was instructed to stand up, and a new 3-minute recording was scheduled.

6.1.2 Initial pre-processing

The first minute of each recording was removed to ensure the analysis was performed on the remaining 2-minute steady-state segment. Each segment was linearly detrended to remove baseline drift and filtered using a fourth-order Butterworth bandpass filter (0.7–12 Hz). This digital filtering was applied to address visible 50 Hz interference, which the analog circuit attenuated but did not entirely eliminate.

6.2 ECG analysis

When estimating the ECG bpm we used the Pan–Tompkins peak detection algorithm with a 8-17 Hz bandpass filter as a default, however we also compared it to other methods like envelope peak detection and a windowed FFT-algorithm. When noticeable discrepancies appeared between the default and other methods, further investigation was required.

6.2.1 ECG investigation and ground truth

For this study we were generally confident in the data quality of the ECG, but given the hardware and the nature of the setup we noticed already during data collection that this assumption did not always hold. If a subject turned their head, moved their body or touched certain objects this caused increased volatility and even complete "blinding" of the signal where the voltage would go to zero or max out at 3.3V.

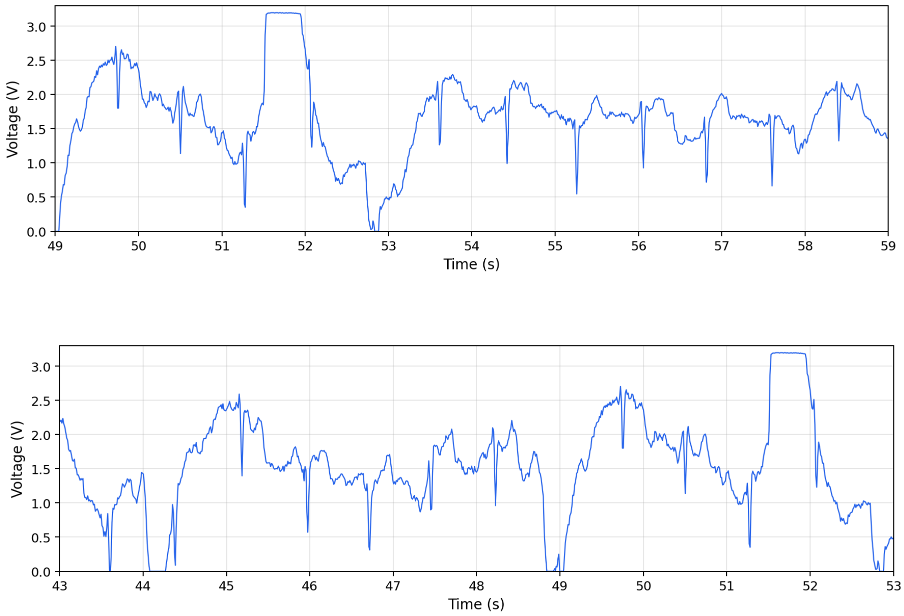

Figure 6.1: Examples of a volatile and blinded intervals (subject 1 standing ECG)

These aberrations naturally revealed themselved during BPM estimation where noticeable discrepancies would arise between the different HR estimation methods. For this reason we would manually annotate peaks in intervals of measurements with high volatility and interpolate the peaks in "blinded" intervals. After this analysis was done, we could have high confidence in the ECG peaks and comfortably rely on them as our ground truth for HR estimates.

6.3 BCG analysis

Before estimating heart rate from the BCG measurements, the signal was digitally filtered.

The circuit end already band limited the signal before digitisation, as shown in Figure 4.1. Our circuit follows the same structure as [6], a charge amplifier, a high pass stage, and a third order Butterworth low pass filter at the end. Digital bandpass filtering was applied nevertheless, because our implementation used different operational amplifiers, different components for data acquisition and because the sensor lead wires picked up visible 50 Hz interference that the analog stage attenuated but did not eliminate.

Each measurement was linearly detrended to remove baseline drift, then filtered with a fourth order Butterworth bandpass between 0.7 and 12 Hz.

6.3.1 BCG heart rate estimation

For BCG measurements, heart rate was estimated from the preprocessed BCG using autocorrelation on the full two-minute segment. The autocorrelation function was evaluated to find the

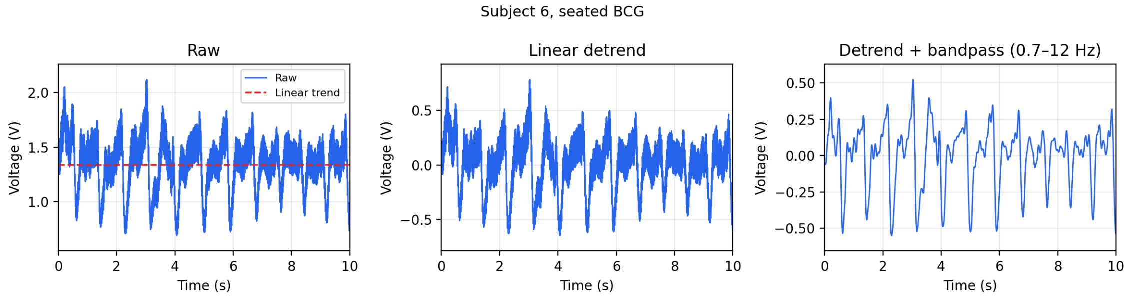

Figure 6.2: Subject 6 seated BCG preprocessing

strongest peak within a targeted lag range of 0.45–1.15 s. This specific band, corresponding to a window of approximately 52 to 133 bpm, was deliberately selected to capture a typical distribution of resting heart rates for subjects in seated or standing positions. If a subject's resting baseline fell significantly outside these bounds this would indicate severe cardiovascular abnormalities such as bradycardia or tachycardia whose detection is beyond the scope of this project. The estimated heart rate was then calculated as $60/\tau$, where τ is the peak lag in seconds.

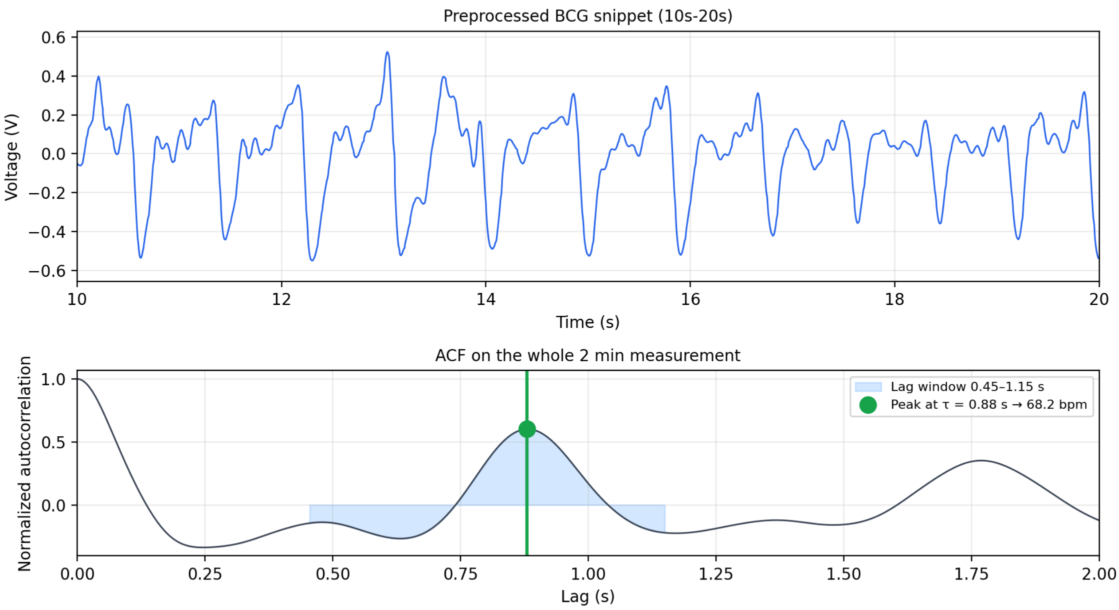

Figure 6.3: Subject 6 preprocessed signal and the autocorrelation function (ACF) for the whole 2 minute measurement.

At the start of each recording we instructed the BCG and ECG computers to begin logging at the same clock time, but startup delay and time drift between the two machines meant we could not rely on subsecond precision between the 2 measurement methods. Peak-to-peak comparison between the ECG and BCG was therefore not used for the analysis of seated recordings. Autocorrelation instead yields one heart rate for the whole segment from the repeating foot pulse pattern, without requiring precise time synchronization.

The CWT-based methodology proposed by [6] was initially explored but proved untenable given the characteristics of the collected data.

Windowed heart rate estimation

The heart rate analysis above yields one heart rate per subject from the full 2 minute measurement. In order to explore an approach with quicker updates, the BCG measurements were also analyzed in fixed 20 second windows.

Each 2 minute segment was split into six consecutive windows. For every window, heart rate was estimated independently. The raw BCG snippet was linearly detrended and bandpass filtered with the same fourth order 0.7–12 Hz Butterworth settings as for the full segment, this conditioning was applied on each window rather than on the complete measurement.

Heart rate from each preprocessed window was then obtained with the same autocorrelation step as above.

Across all 13 subjects this produced 78 windows for each setting.

7 Results and evaluation

This chapter presents the experimental findings for the insole-based BCG system in seated and standing conditions, using 2-minute recordings and 20-second windows.

7.1 Results

Tables 7.1 and 7.2 summarize per-subject heart rate from ECG and BCG (ACF). Bold values in the standing table denote deviations within ± 5 bpm.

Table 7.1: Sitting heart rate per subject and seated BCG HR 20 s window evaluation.

	Overall Sitting HR			20 s Windows	
ID	ECG	BCG (ACF)	$\lvert\Delta\text{bpm}\rvert$	Within ± 5 bpm	Mean $\lvert\Delta\text{bpm}\rvert$
1	75.51	76.91	1.40	6/6	1.24
2	94.59	94.49	0.10	5/6	5.74
3	75.33	74.55	0.78	6/6	0.75
4	75.90	75.11	0.79	6/6	1.01
5	59.06	60.64	1.58	5/6	14.06
6	68.77	68.20	0.57	6/6	0.93
7	85.11	85.73	0.62	6/6	1.95
8	98.39	98.42	0.03	6/6	0.43
9	55.80	56.09	0.29	5/6	3.18
10	86.20	86.33	0.13	4/6	9.90
11	61.40	60.02	1.38	6/6	1.06
12	68.45	65.93	2.52	3/6	4.76
13	87.48	85.77	1.71	6/6	2.01
Average	**76.31**		**0.91**	**89.7%**	**3.62**

Table 7.2: Standing heart rate per subject and standing BCG HR 20 s window evaluation.

	Overall Standing HR			20 s Windows	
ID	ECG	BCG (ACF)	$\lvert\Delta\text{bpm}\rvert$	Within ± 5 bpm	Mean $\lvert\Delta\text{bpm}\rvert$
1	84.24	63.56	20.68	0/6	20.79
2	97.52	95.20	**2.32**	4/6	6.61
3	88.95	67.43	21.52	0/6	25.92
4	89.52	90.39	**0.87**	3/6	8.07
5	67.33	71.87	**4.54**	1/6	12.17
6	76.12	54.05	22.07	1/6	11.73
7	94.90	65.93	28.97	0/6	31.77
8	103.44	55.85	47.59	0/6	44.18
9	62.92	61.84	**1.08**	3/6	6.05
10	97.56	60.93	36.63	0/6	32.76
11	78.19	73.63	**4.56**	2/6	9.75
12	82.78	63.86	18.92	2/6	13.38
13	93.98	58.31	35.67	1/6	27.91
Average	**85.96**		**18.88**	**21.8%**	**19.32**

7.2 Evaluation

7.2.1 Seated BCG

As we can see the average deviation from the ECG is 0.91 bpm which is well within the AAMI deviation limits of ± 5 bpm. Even the highest deviation 2.52 bpm clears this threshold handily. Below is a Bland-Altman plot for the 2 minute seated table. [31].

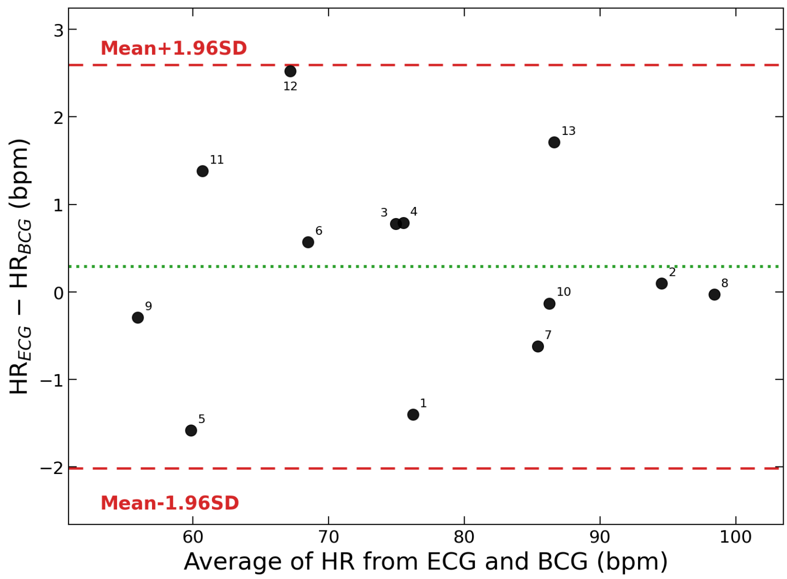

Figure 7.1: Bland-Altman plot for the 2 minute seated measurements

20 s intervals

For the 20s interval analysis we see a reliable HR estimate in almost 90% of cases, showing promise for quick HR adjustments with the current approach.
The mean absolute deviations for this analysis are typically dominated by 1 or 2 big misses as can be seen in Table A.1.

Variability in measurement quality

While conducting the seated BCG measurements it was noted that there was clear variability in the quality of the measurements. While some measurements had little to no noise and traced with military precision a repetitive pattern, others had a much more convoluted structure. To illustrate this we have added 3, 10-second snippets of raw data below:

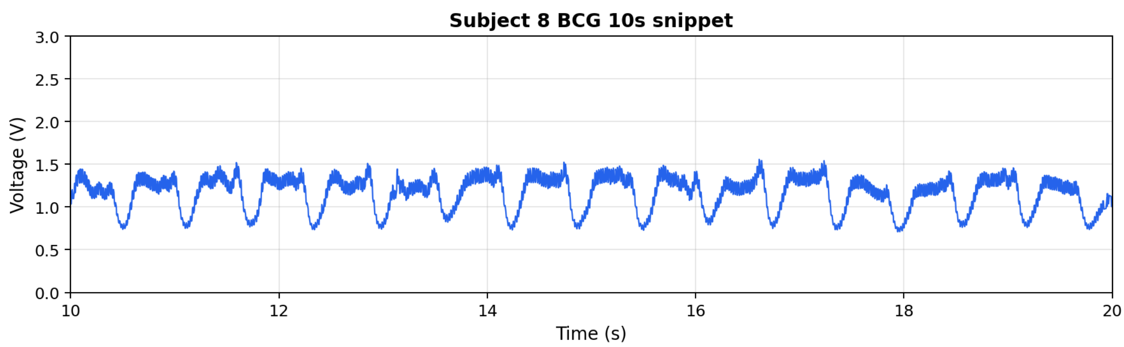

Figure 7.2: Example of a low-noise seated BCG measurement

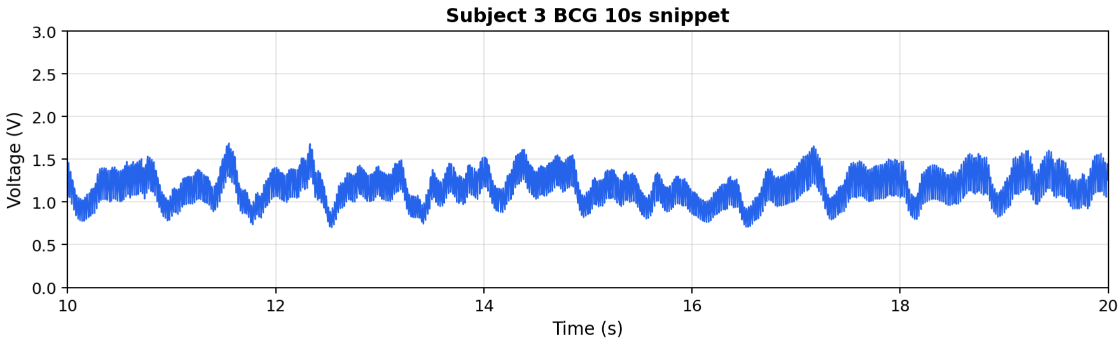

Figure 7.3: Example of a noisy seated BCG measurement

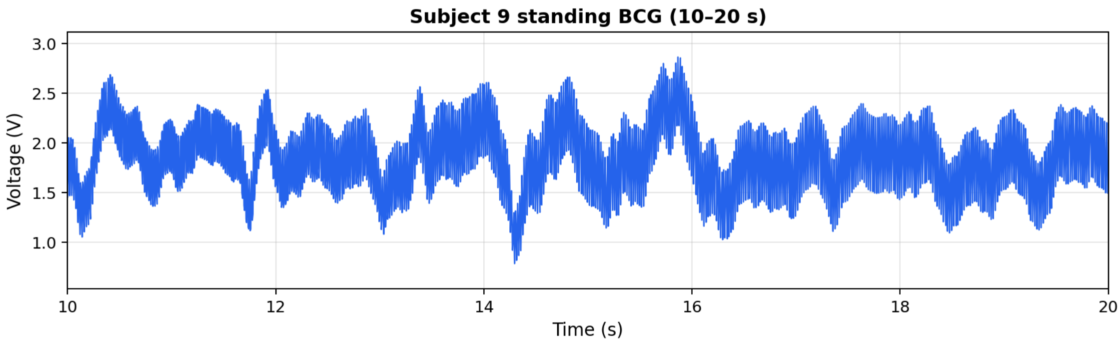

Figure 7.4: Example of a standing BCG measurement with clear 50 Hz noise

As we can see, the measurement for subject 8 has significantly less noise, especially at the "troughs" where the signal falls and rises drastically for a short period of time.
Conversely, the measurement for subject 3 has clear 50 Hz noise attached to the graph along the whole interval. The signal seems much less repetitive and requires a generous eye to classify the sharp, downward slopes as heartbeats. The standing measurement clearly shows a higher level of 50 Hz noise being picked up which was generally observed in this setting.

This difference in measurement quality shows up in the gap between ECG and BCG heart rate. For the seated examples, subject 8 had the lowest deviation (0.03 bpm) and subject 3 had 0.78 bpm. For the standing example, subject 9 had a deviation of 1.08 bpm.

For the 20 second windows, an indirect comparison can be made with [6]. They achieved a coverage factor of 95.6%, we were able to estimate heart rate within the AAMI limits in 89.7% of the windows.

7.2.2 Standing BCG

Table 7.2 shows large deviations between trusted ECG and BCG for standing measurements. The mean absolute difference was 18.88 bpm, while five of thirteen subjects were within the ± 5 bpm band. Errors ranged from 0.87 to 47.59 bpm.
A similar 20s window analysis for standing shows low reliability with only 21.8% of the intervals being within the ± 5 bpm band.
To illustrate part of the challenge with standing measurements, several standing BCG segments are shown below with volatile data:

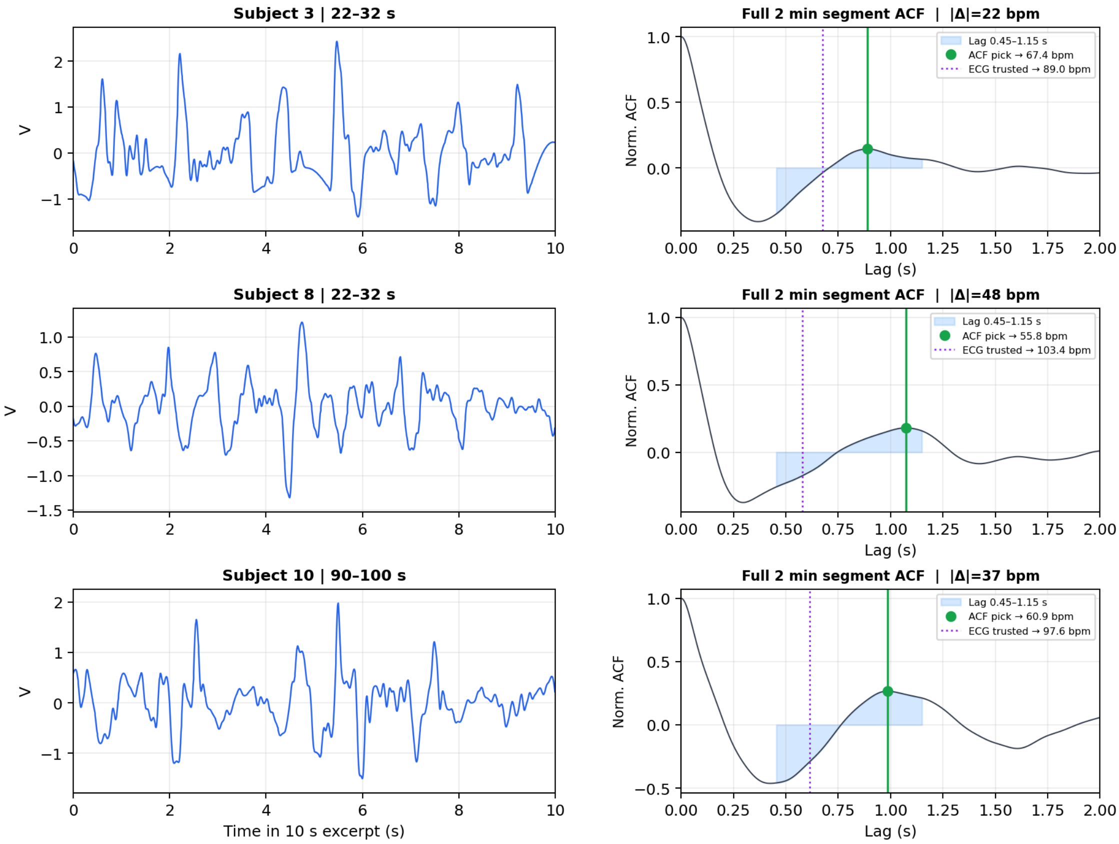

Figure 7.5: Examples of noisy intervals in standing preprocessed BCG measurements along with their full 2 minute autocorrelation function (ACF)

We strongly believe much of this volatility is caused by motion artifacts, such as a subject rebalancing, moving their foot slightly or another motion altogether.

8 Discussion

This chapter evaluates the outcomes of our feasibility study, reflecting on the system's limitations and the impact of our hardware design choices. We analyze the vulnerabilities of our setup, the electrical interference encountered in the laboratory environment, and how these factors compounded to degrade the signal.

8.1 Hardware Integration and Mechanical Vulnerabilities

The analog front-end was exclusively developed and maintained on a breadboard. While this enabled quicker prototyping and component swapping, using a perfboard would undoubtedly have streamlined the circuit and let us solder the wire ends of the piezoelectric sensor to the actual circuit minimizing exposed surface area. Because every component was held solely by the breadboard's contacts in our circuit, the circuit was fragile and sensitive to external movement. Another possibility could have been using a PCB which would have provided robust shielding but would have required significantly more time and expertise.

This vulnerability of our circuit was also accentuated by our method of sensor integration. Rather than utilizing robust, shielded connectors, the bare wire ends of the piezoelectric sensor were manually inserted into the breadboard. When a subject measurement was finished, they would be pulled out again. Repeated insertion, extraction, and physical handling during testing degraded the delicate wire ends. To maintain a functional connection, we frequently had to cut, strip, and re-solder the tips of the sensor wires. This physical degradation resulted in varying lengths of exposed, unshielded wire directly at the input of the charge amplifier, contributing to the variability in measurement quality.

8.2 Electromagnetic Interference

The exposed piezoelectric wire leads acted as highly effective antennas, picking up the ambient electromagnetic interference present in the environment. The measurements were conducted in an electronics lab, an environment with active computers, power supplies, and other instrumentation. This resulted in significant 50 Hz mains hum present in the raw signal.

8.3 Component trade-offs

In our circuit we substituted the AD8641 and OP207 amplifiers used in previous literature with the MCP6002 for the first two stages [6]. It is at the high-impedance front end that this substitution carries the most weight. The charge amplifier interfaces with the piezoelectric sensor through a $\approx 3\,\mathrm{G\Omega}$ T-network. At this node, the relevant parameter is the input bias current—the small DC current drawn by the op-amp's input stage, which has nowhere to go but through the feedback network. Across gigaohms, even a few picoamps develop a substantial DC offset and drift.

This is why García-Limón et al. used the AD8641, a JFET-input part specifying 0.25 pA typical and 1 pA maximum bias current. The MCP6002 we substituted specifies 1 pA typical but 19 pA maximum [32, 33], a roughly nineteen-fold increase in the worst case.

8.3.1 Parasitic Capacitance

Solderless breadboards inevitably introduce parasitic capacitance. Their parallel metal strips, separated by a plastic dielectric, form unintended capacitors between adjacent rows (typically around 2-5 pF).

While negligible in many standard applications, this is a worthwhile consideration for high-impedance front-ends. Our charge amplifier relies on a T-network with a massive equivalent resistance of $\approx 3\,\text{G}\Omega$. A capacitor blocks low-frequency signals but offers progressively less resistance to higher frequencies.

At our target heartbeat frequency of roughly 1 Hz, the breadboard's parasitic capacitance provides high resistance, ensuring the cardiac signal travels correctly through our intended $3\,\text{G}\Omega$ network. However, at higher frequencies such as 50 Hz, this parasitic path becomes slightly more conductive. As a result, the breadboard's internal structure creates a minor secondary path for high-frequency signals. While not severe enough to completely bypass our filtering, it does allow a small amount of 50 Hz environmental noise to slowly couple across the board, marginally contributing to the baseline interference detailed in Section 8.2.

8.4 Interpreting the results

The hardware issues previously discussed explain the 50 Hz noise in our raw data. The unshielded breadboard and exposed sensor wires picked up electrical interference from the room. This interference partly polluted the signal before we could perform out digital filtering, meaning the data used for HR estimation degraded from the outset.

Despite this noisy baseline, we still got reliable heart rates when subjects were sitting down. As Table 7.1 shows, the average absolute deviation from the reference was only 0.91 bpm. This worked because a seated person is relatively still, making it easier to isolate the BCG recoil forces.

8.4.1 Data analysis choices

We decided to use autocorrelation instead of other methods like the Fast Fourier Transform (FFT). As Figure 8.1 shows, the FFT often got confused by the noisy data. Its highest peak would sometimes guess exactly double the actual heart rate (like 120 bpm instead of 60 bpm). Because a doubled rate is still a realistic human heart rate, we couldn't rule it out as a priority. Autocorrelation avoids this trap because it looks for the entire repeating pattern of the heartbeat, finding the true rhythm even in messy data.

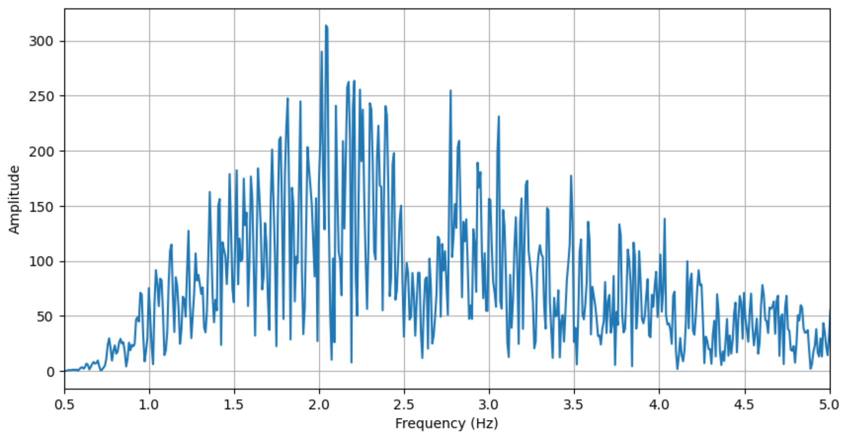

Figure 8.1: A FFT of a pre-processed sitting measurement demonstrating harmonic ambiguity (reference ECG shows HR of 59.1 bpm $\approx$ 0.985 Hz).

We also tested the algorithm on shorter 20-second windows to see if a real-time prototype could give quick updates. Here, 89.7% of the estimates were within the AAMI limit of ±5 bpm. This is worse than the analogous coverage factor from García-Limón et al. [6] at 95.6%.

The standing measurements clearly showed the limits of our setup, leaving a massive average error of 18.88 bpm. A closer look at Table 7.2 generally shows underestimation of the HR with results such as the subject 8 estimates implying 50 % undercounting (ECG - 103.44 bpm, BCG - 55.85 bpm).

8.4.2 Result limitations

In general there were 2 metrics within the scope of this project which we were not able to measure. The first one was the time between R-peak to J-peak time delay. The reason for this was the way we initiated the data capture on the 2 PCs. Data capture runs were scheduled for a specific clock time but due to time drift of the 2 computers and difference in startup time we could not rely on this simultaneous start at a subsecond precision level.
The second metric was the identification of J peaks and J-to-J time difference. These metrics were identified in [6] but due to the data limitations previously established we were not able to replicate their data cleaning pipeline and ultimately reliable J-peak detection was not possible with our setup.

9 Future work

This chapter outlines recommended directions for future iterations of this project, ranging from methodological to computational considerations. We outline potential improvements for cohort diversity and analyze the viability of digitally processing the signal on a smaller device.

9.1 Methodological

A larger sample size would be of considerable benefit. The original study recruited 25 volunteers, a scale this work was unable to match. A more pressing limitation than sample size, however, is its lack of diversity: our cohort consisted almost entirely of healthy young men. The performance of the system on women, children, elderly individuals, and those with underlying health conditions therefore remains unknown. Because the BCG signal depends strongly on body composition, the circuit cannot be claimed to generalize across populations without substantially more testing.

A dedicated, electrically clean, and private room for the experiments would also help attenuate 50 Hz interference and would simplify scheduling, in turn increasing the achievable sample size. Such a setting would also address subject comfort: some volunteers may be reluctant to be measured in a semi-public space, and a more private environment could encourage participation from women and individuals of differing body types, thereby improving the diversity of the sample.

Extending the measurements to walking subjects is a further possibility. Although standing trials were conducted, the present setup cannot demonstrate the circuit's efficacy on a walking subject. The signal would be expected to be considerably harder to extract, since standing alone already introduces substantial noise; a sharper filter could be one way forward. This would be a longer-term objective, as it first requires a more mechanically stable circuit: the current setup is sensitive even to slight movements of the foot, and a single step would be enough to disrupt it entirely.

As discussed in the interpretation section (Section 8.4), there is reason to believe that the signal observed while standing corresponded to the subject's postural sway rather than the heartbeat. This hypothesis could be tested directly by adding an accelerometer to measure the sway. Two outcomes are possible: either the sway occupies frequencies below the heart rate (below ~0.7 Hz), or it overlaps with it. In the former case, a highpass filter with a corner near 0.6 Hz, above the sway but below the slowest expected heart rate, would remove the sway while preserving the heartbeat. In the latter case, no fixed filter can separate the two components, as they share the same frequency band; this would demand further analysis.

9.2 Additional cardiovascular metrics

While this study focused primarily on estimating an average heart rate over a given window, the BCG signal contains additional information that could be used for determining other cardiovascular metrics.

Recently, García-Limón et al. [28] demonstrated that insole-based BCG can successfully estimate Heart Rate Variability (HRV). Building on this, future iterations of our system could be tested on subjects with known cardiac conditions to see if beat-to-beat HRV tracking can reliably detect arrhythmias.

Another candidate for future work in this area would be identification of Atrial Fibrillation (AFib), an irregular rhythm that is often intermittent and easily missed during brief doctor visits. As highlighted by the massive success of the Apple Heart Study [34], everyday consumer wearables possess immense potential to catch these hidden conditions. If identification proves viable on an insole, it could offer a similarly unobtrusive way to continuously monitor for AFib during a user's normal daily routine.

9.3 On-device analysis

One of the most promising avenues for future work would be to perform the analysis on-board using an embedded system. Such a system would acquire the BCG data from the analog front-end and then filter and process it on-device. The feasibility of running biometric algorithms directly on shoe-mounted microcontrollers has already been demonstrated for plantar biometrics using edge computing [35].

Our autocorrelation algorithm samples at 200 Hz, and only lags within the plausible heart-rate range (roughly 0.45–1.15 s, a span of 0.7 s) need to be evaluated. At 200 Hz this corresponds to approximately 140 lags rather than the full window. A 20-second window contains 4000 samples, each correlated against these 140 lags, giving a total workload per window of

$$4000 \cdot 140 = 560{,}000 \text{ \textit{MACs (multiply–accumulate operations)}}$$

Dividing by the 20-second window length gives

$$\frac{560{,}000}{20} = 28{,}000 \text{ \textit{MACs/s}}$$

Even under pessimistic assumptions, this load is well within the capabilities of most microcontrollers [36] [37].

The more significant bottleneck is memory. The entire measurement is not stored, but a full 20-second window must be retained in order to correlate it against itself. The ADC produces a 14-bit sample, but each is stored in a 16-bit word, so the window requires

$$4000 \cdot 16 = 64{,}000 \text{ bits}$$

$$\frac{64{,}000 \text{ bits}}{8} = 8 \text{ kilobytes}$$

This makes the choice of microcontroller less clear-cut, but suitable candidates remain, such as the nRF52832 or the arduino R4 minima [38, 39], which provide suitable amounts of RAM.

10 Conclusion

This work set out to investigate whether heart rate could be estimated from an insole-based ballistocardiography system, using piezoelectric sensors and relatively cheap consumer-grade hardware rather than specialized laboratory equipment. In this sense the project was partly successful.

For the seated position, the system was able to estimate BPM over two minute intervals with an average absolute deviation of 0.91 bpm from the ECG reference. Even the worst seated estimate was within 2.52 bpm, and therefore inside the AAMI limit of ± 5 bpm. When the same seated measurements were split into 20 second windows, 89.7% of the windows were still within the same AAMI limit. This shows that cheap piezoelectric smart-insoles can be used to extract useful heart rate estimates when the subject is seated and reasonably still.

We were not able to reliably estimate BPM in the standing position. 5 out of 13 measurements gave reasonable estimates, but in general the standing signal was dominated by motion artifacts, and the average error was far outside the desired range.

In conclusion, our results support the idea that insole-based BCG is a viable direction for future development and research.

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A Appendix

A.1 Consent form

This form asks for your consent to participate in a technical feasibility study that investigates whether heart rate can be estimated from signals recorded at the foot using a piezoelectric sensor embedded in an insole. During the study, the research team will ask you to place your foot on the instrumented insole while measurements are taken. We will also record a reference electrocardiogram (ECG) at the same time. The study is for research purposes only and is not intended for diagnosis, treatment, or medical advice.

If you agree to participate, DTU will collect the signal recorded from the insole together with the ECG recording, your age, and your height.

Participation is voluntary. You may decline to participate or stop the measurement at any time without giving a reason and without any negative consequences.

Your study data will be handled confidentially and stored only for the duration necessary to complete the analysis. All collected study data associated with you will be deleted no later than six months after collection unless a longer retention period is required by law or unless you separately agree in writing to extended retention.

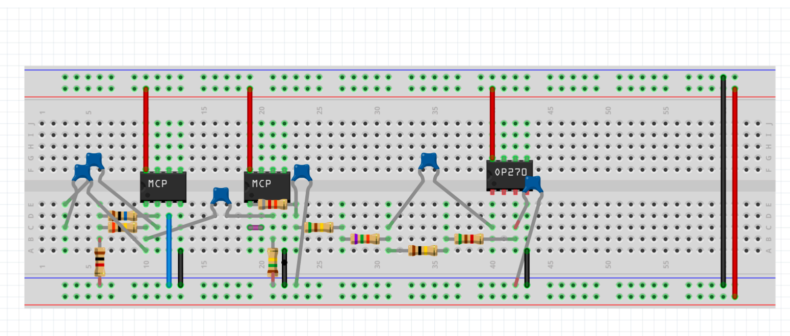
By signing below, you confirm that you are at least 18 years old, that you have read and understood this form, that you have had the opportunity to ask questions, and that you voluntarily consent to participate in this study.

Participant name: ______________________________

Signature: ______________________________

Date: ______________

A.2 Breadboard wiring



A.3 Supplementary values for seated BCG 20 s intervals

Table A.1: Seated BCG: absolute heart rate error per 20 s window (ECG vs ACF).

ID	0–20 s	20–40 s	40–60 s	60–80 s	80–100 s	100–120 s	Mean $\lvert\Delta\rvert$
1	0.10	1.68	0.51	1.31	3.11	0.74	1.24
2	0.41	0.96	**30.89**	0.25	0.23	1.72	**5.74**
3	0.19	0.67	2.50	0.44	0.44	0.28	0.75
4	1.75	0.20	0.32	0.58	0.40	2.82	1.01
5	0.74	0.59	3.57	3.22	**74.39**	1.86	**14.06**
6	2.29	0.95	0.72	0.30	0.13	1.18	0.93
7	2.77	3.39	1.30	0.67	1.71	1.83	1.95
8	0.40	0.45	0.89	0.30	0.15	0.40	0.43
9	1.92	1.39	1.49	0.10	0.26	**13.91**	3.18
10	**28.88**	0.25	0.41	**28.65**	1.13	0.10	**9.90**
11	0.20	0.99	4.38	0.40	0.35	0.05	1.06
12	**6.14**	0.57	**5.70**	**12.78**	1.50	1.88	4.76
13	1.44	0.43	3.87	2.49	3.58	0.22	2.01
Average							**3.62**

A.4 Arduino measurement code

```
#include <Arduino.h>

const unsigned long SAMPLE_HZ = 200;
const unsigned long SAMPLE_INTERVAL_US = 1000000UL / SAMPLE_HZ;
const int ADC_BITS = 14;
const int ADC_MAX = (1 << ADC_BITS) - 1;
const float ADC_REF_VOLTS = 5.0f;

void setup() {
  Serial.begin(115200);
  analogReadResolution(ADC_BITS);
}

void loop() {
  static unsigned long nextTick = micros();

  unsigned long now = micros();
  if ((long)(now - nextTick) < 0) {
    return;
  }
  nextTick += SAMPLE_INTERVAL_US;

  int r0 = analogRead(A0);
  float v0 = (r0 / (float)ADC_MAX) * ADC_REF_VOLTS;

  Min, Max, CurrentValue
  Serial.print(0.0f, 4);
  Serial.print(',');
  Serial.print(7.0f, 4);
  Serial.print(',');
  Serial.println(v0, 4);
}
```

A.5 BCG heart rate functions

```python
import numpy as np
from scipy import signal
from scipy.signal import butter, sosfiltfilt

LAG_LO, LAG_HI = 0.45, 1.15


def preprocess_bcg(voltage: np.ndarray, fs: float) -> np.ndarray:
    v = signal.detrend(np.asarray(voltage, dtype=float), type="linear")
    nyq = fs / 2.0
    hi = min(12.0, 0.45 * nyq)
    sos = butter(4, [0.7, hi], btype="band", fs=fs, output="sos")
    return sosfiltfilt(sos, v)


def autocorrelation(x: np.ndarray) -> np.ndarray:
    x = np.asarray(x, dtype=float) - np.mean(x)
    n = len(x)
    nfft = 1 << (2 * n - 2).bit_length()
    spectrum = np.fft.rfft(x, n=nfft)
    ac = np.fft.irfft(spectrum * np.conj(spectrum), n=nfft)[:n].real
    if ac[0] != 0:
        ac /= ac[0]
    return ac

def acf_bpm(x: np.ndarray, fs: float) -> float:
    ac = autocorrelation(x)
    lags = np.arange(len(ac)) / fs
    mask = (lags >= LAG_LO) & (lags <= LAG_HI)
    if not np.any(mask):
        return float("nan")
    k = np.argmax(ac[mask])
    tau = lags[mask][k]
    return 60.0 / tau


def estimate_hr(time_s: np.ndarray, voltage: np.ndarray) -> float:
    dt = np.diff(time_s)
    fs = 1.0 / np.median(dt[dt > 0])
    x = preprocess_bcg(voltage, fs)
    return acf_bpm(x, fs)
```

Listing A.1: BCG preprocessing and autocorrelation HR estimation (bcg_hr_analysis.py).

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